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On convergence rates of Kaczmarz-type methods with different selection rules of working rows ☆

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ABSTRACT

The Kaczmarz method is a classical while effective iteration method for solving very large-scale consistent systems of linear equations, and the randomized Kaczmarz method is an important and valuable variant of the Kaczmarz method. By theoretically analyzing and numerically experimenting several criteria typically adopted in the non-randomized and the randomized Kaczmarz method for selecting the working row, we derive sharper upper bounds for the convergence rates of some of the correspondingly induced Kaczmarz-type methods including those with respect to the maximal residual, maximal distance, and distance selection rules of the working row, and, for this whole suite of iteration methods consisting of the Kaczmarz methods with respect to the uniform, non-uniform, residual, distance, maximal residual, and maximal distance selection rules of the working row, we reveal their comparable relationships in terms of both mean-squared distance and mean-squared error, and show their computational effectiveness and numerical robustness based upon implementing a large number of test examples. Here the mean-squared distance is defined as the mean-value of the squared Euclidean norm of the current update increment of the iteration, and the mean-squared error is defined as the mean-value of the squared Euclidean norm of the current error that is the difference between the current iterate and the true solution of the target linear system.

1. Introduction

We consider iterative solution of the large-scale consistent system of linear equations

$$Ax = b, \quad \text{with} \quad A \in \mathbb{C}^{m\times n} \quad \text{and} \quad b \in \mathcal{R}(A). \tag{1.1}$$

More specifically, here $A \in \mathbb{C}^{m\times n}$ is a complex $m \times n$ matrix, $b \in \mathbb{C}^m$ is a complex m-dimensional vector belonging to the range space $\mathcal{R}(A)$ of the coefficient matrix $A \in \mathbb{C}^{m\times n}$, and $x \in \mathbb{C}^n$ is a complex n-dimensional unknown vector, with $\mathbb{C}^m$ and $\mathbb{C}^{m\times n}$ being the complex vector and the complex matrix space of the dimensions m and $m \times n$, respectively.

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The Kaczmarz method [21], originally proposed by the Polish mathematician Stefan Kaczmarz in 1937, is a classical while simple and effective iteration method for solving the consistent linear system (1.1). In many fields such as image restoration, image reconstruction, computerized tomography, and digital signal processing, the Kaczmarz method is also popularly known as the *algebraic reconstruction technique* (**ART**). In this method, at each iteration step k, only one row, say, e.g., i_k, is chosen from the coefficient matrix $A \in \mathbb{C}^{m\times n}$; this kind of selected row is named as the working row. Then the current iterate, denoted as x_k, is projected onto the corresponding i_k-th hyperplane described by the equation $A^{(i_k)}x = b^{(i_k)}$, so that the next iterate x_{k+1} is well-defined as the resulting projection point. Here $A^{(i_k)}$ and $b^{(i_k)}$ denote the i_k-th row of the matrix $A \in \mathbb{C}^{m\times n}$ and the i_k-th entry of the vector $b \in \mathbb{C}^m$, respectively. A great quantity of practical applications has exhibited that sweeping through the rows of the matrix $A \in \mathbb{C}^{m\times n}$ randomly rather than sequentially can greatly improve the convergence speed of the Kaczmarz method. This phenomenon may be caused mainly by an undesirable ordering of the rows of the coefficient matrix $A \in \mathbb{C}^{m\times n}$ or, in other words, by an unfavorable arrangement of the equations in the linear system (1.1). In consideration of this fact, Strohmer and Vershynin [36] proposed the *randomized Kaczmarz* (**RK**) method in 2009, and proved its expected exponential rate of convergence or the "linear convergence rate". Afterwards, theoretical analysis, algorithmic development, and numerical validation about the RK method have been further advanced, deepened, and completed. For more details, we refer to [3–12,16,20,24,26–31,35,40,42] and the references therein. In addition, we refer to [22,23,25,33,34,37–39] and the references therein for advanced and widespread applications of the RK method in the fields such as magnetic particle imaging, learning theory, massive MIMO systems, phase retrieval, compressed sensing, network tomography, and ill-posed problems.

In the original RK method proposed in [36], the working row is selected from the coefficient matrix $A \in \mathbb{C}^{m\times n}$ with the probability being proportional to the squared Euclidean norms of its rows. This selection strategy of the working row is called a non-uniform selection rule, which degenerates to the so-called uniform selection rule if all rows of the matrix $A \in \mathbb{C}^{m\times n}$ happen to have the same Euclidean norm, resulting in a probability exactly equal to $\frac{1}{m}$. Alternatively, the working row can be also selected from the coefficient matrix $A \in \mathbb{C}^{m\times n}$ with the probability being equal to the square of the ratio of the entry to the Euclidean norm of either the current residual vector or the current update-increment (or distance) vector, which are termed as the residual selection rule or the distance selection rule, correspondingly; see [18,19]. Besides, there are also discussions that explore applications of the greedy selection rule in the Kaczmarz method. For example, the maximal residual selection rule [17] and the maximal distance selection rule [15,41], etc.

Though many different selection rules have been proposed and employed in various non-randomized or randomized variants of the Kaczmarz method, yet there are quite few results in comparing and examining advantages and disadvantages of these Kaczmarz-type methods in the viewpoint of either theory or computations. Here it is worthwhile to state that in 2016 Nutini, Sepehry, and Laradji, et al., provided a few comparison results about the Kaczmarz-type methods with respect to the uniform, non-uniform, maximal residual, and maximal distance selection rules of the working row; see [32]. We clarify that the distance is defined as the Euclidean norm of the difference between the current iterate and its previous adjacent iterate, i.e., the current update increment of the iteration. From this work, we have learned that the Kaczmarz method using the maximal distance selection rule presents the smallest upper bound of the convergence rate and shows the best numerical performance in terms of the elapsed computing time, while the relationships among the other three selection rules are non-deterministic and, in general, problem-dependent too, in terms of both theoretical property and numerical behavior.

In this paper, we continue the research work in [32] to further study the theoretical properties and numerical behaviors of the above-mentioned four Kaczmarz-type methods that use the uniform, non-uniform, maximal residual, and maximal distance selection rules of the working row, together with another two randomized Kaczmarz methods that use the residual and the distance selection rule of the working row. More concretely, these six Kaczmarz-type methods are the Kaczmarz methods with respect to the uniform, non-uniform, residual, distance, maximal residual, and maximal distance selection rules of the working row, and we abbreviate them as RKU, RKNU, RKR, RKD, KMR, and KMD, respectively; see Tables 1 and 2. Note that the former four are randomized iteration methods, while the latter two are non-randomized iteration methods. In addition, if all rows of the coefficient matrix $A \in \mathbb{C}^{m\times n}$ of the linear system (1.1) have the same norm, then the RKNU, RKD, and KMD methods degenerate, respectively, to the RKU, RKR, and KMR methods. Our comparability study is mainly focused on the mean-squared distance and on the upper bound for the convergence factor of the mean-squared error, which are derived for measuring the convergence rates of the Kaczmarz-type methods from different aspects.

In theory, we have demonstrated that the mean-squared distance and the upper bound for the convergence factor of the KMD method are the smallest among these six Kaczmarz-type methods. In particular, these two kinds of measuring quantities with respect to the RKD method are smaller than those with respect to the RKU method, and these measuring quantities of the RKR method are smaller than those of the RKNU method. Besides, we give specific examples to illustrate and confirm that the other mutual relationships among the afore-mentioned six Kaczmarz-type methods are incomparable and problem-dependent.

In computations, we have observed that the KMD method shows the best performance among these six Kaczmarz-type methods. In particular, the RKD and RKR methods perform better than both RKU and RKNU methods. Also, we have examined the other non-deterministic relationships among the convergence properties of these Kaczmarz-type methods, which have been explored from and explained by the above-mentioned specific examples. It has been shown that these incomparable relationships are indeed problem-dependent. In addition, when a few rows of the coefficient matrix $A \in \mathbb{C}^{m\times n}$ have norms far greater than the other rows, the RKU method exhibits much better numerical behavior than the RKNU

Table 1
The abbreviations for Kaczmarz-type methods.

Method	Description
RKU	randomized Kaczmarz with uniform rule [36]
RKNU	randomized Kaczmarz with non-uniform rule [36]
RKR	randomized Kaczmarz with residual rule [6,7,19]
RKD	randomized Kaczmarz with distance rule [16]
KMR	Kaczmarz with maximal residual rule [17]
KMD	Kaczmarz with maximal distance rule [7,15,41]

Table 2
The abbreviations for selection rules of working rows.

Rule	Description
U	uniform rule [36]
NU	non-uniform rule [36]
R	residual rule [6,7,19]
D	distance rule [16]
MR	maximal residual rule [17]
MD	maximal distance rule [7,15,41]

method. Consequently, with numerical implementations of the Kaczmarz-type methods on a large set of vast test problems, we have shown that the experimental results nicely coincide with the theoretical results.

The organization of the paper is as follows. In Section 2, we introduce the six Kaczmarz-type methods. In Section 3, we give the upper bounds for the convergence rates of these methods and present certain comparable results from the two aspects stated previously. In Section 4, we use examples to illustrate the other incomparable relationships among these methods. The experimental results and comparability discussions are reported in Section 5. Finally, in Section 6, we end the paper with conclusions and remarks.

At the end of this section, we precisely describe the notation adopted in this paper. For a complex vector q, we use q^T, q^*, $\|q\|$, and $\|q\|_\infty$ to denote its transpose, conjugate transpose, Euclidean norm, and infinity norm, respectively. And for a complex matrix G, we use G^T, G^*, $\|G\|_F$, $\mathcal{R}(G)$, and $\sigma_{\min}(G)$ to indicate its transpose, conjugate transpose, Frobenius norm, range space, and minimal nonzero singular value, respectively. In addition, $G^{(i)}$ represents the i-th row of the matrix G, and $q^{(i)}$ represents the i-th entry of the column vector q. In particular, $\mathbf{1}_m$ is used to denote the m-dimensional column vector with all entries being equal to 1, and I_m is employed to denote the m-dimensional identity matrix. Besides, we use $\mathbb{E}\cdot$ to indicate the mean-value of the corresponding random variable. For more details, we refer to [2,6] and the references therein.

2. The Kaczmarz-type methods

In this section, we give brief and precise descriptions about the above-stated Kaczmarz-type methods and the associated selection rules of the working row, including the randomized Kaczmarz method and its randomized selection rule.

In the sequel, without loss of generality, we assume that the coefficient matrix $A \in \mathbb{C}^{m\times n}$ of the linear system (1.1) does not have a zero row.

We first precisely restate the RK method in its general form as Method 1.

Method 1 (The RK method).

Input: A, b, ℓ, p_i $(i=1,2,\ldots,m)$, and x_0
Output: x_ℓ
1: **for** $k=0,1,\ldots,\ell-1$ **do**
2: Select $i_k \in \{1,2,\ldots,m\}$ with probability $\Pr(\text{row}=i_k)=p_{i_k}$
3: Set $x_{k+1}=x_k+\frac{b^{(i_k)}-A^{(i_k)}x_k}{\|A^{(i_k)}\|^2}\left(A^{(i_k)}\right)^*$
4: **endfor**

The selection rules of the working row in the RK method, that is, in Method 1, are totally determined by the probability criterions adopted in Step 2, and different selection rules can result in different Kaczmarz-type methods. In particular, if the selection rule is deterministic, then the randomized Kaczmarz method degenerates to the non-randomized Kaczmarz method and, moreover, if the working row is selected in order cyclically from the index set $\{1,2,\ldots,m\}$ so that $i_k=(k \bmod m)+1$, then the randomized Kaczmarz method becomes the classical Kaczmarz method.

The most typical and representative selection rules of the working row in the Kaczmarz-type methods are listed as follows; see Tables 1 and 2.

Table 3
Computational costs of the k-th iterate for the Kaczmarz-type methods.

Method	Initialization	Arithmetic Operations	Maximum Operations
RKU	$2\,\mathrm{nnz} - m$	$4\,\mathrm{nnz}_{i_k} + 1$	0
RKNU	$2\,\mathrm{nnz} - m$	$4\,\mathrm{nnz}_{i_k} + 1$	0
RKR	$2\,\mathrm{nnz} - m$	$2\,\mathrm{nnz} + 2\,\mathrm{nnz}_{i_k} + m + 1$	0
RKD	$2\,\mathrm{nnz} - m$	$2\,\mathrm{nnz} + 2\,\mathrm{nnz}_{i_k} + 2m + 1$	0
KMR	$2\,\mathrm{nnz} - m$	$2\,\mathrm{nnz} + 2\,\mathrm{nnz}_{i_k} + m + 1$	χ_m
KMD	$2\,\mathrm{nnz} - m$	$2\,\mathrm{nnz} + 2\,\mathrm{nnz}_{i_k} + 2m + 1$	χ_m

- The uniform selection rule or the U rule: $p_{i_k} = \frac{1}{m}$, which leads to the RKU method.
- The non-uniform selection rule or the NU rule: $p_{i_k} = \frac{\|A^{(i_k)}\|^2}{\|A\|_F^2}$, which leads to the RKNU method.
- The residual selection rule or the R rule: $p_{i_k} = \frac{|r_k^{(i_k)}|^2}{\|r_k\|^2}$, which results in the RKR method. Here $r_k = b - Ax_k$ is the residual vector with respect to the k-th iterate x_k, and $r_k^{(i_k)}$ is its i_k-th entry.
- The distance selection rule or the D rule: $p_{i_k} = \frac{\left(d_k^{(i_k)}\right)^2}{\|d_k\|^2}$, which results in the RKD method. Here d_k is the distance vector with its i-th entry being defined as $d_k^{(i)} = \frac{|r_k^{(i)}|}{\|A^{(i)}\|}$ for $i = 1, 2, \ldots, m$.
- The maximal residual selection rule or the MR rule: $i_k = \underset{1\le i\le m}{\operatorname{argmax}}\ |r_k^{(i)}|$, which gives the KMR method.
- The maximal distance selection rule or the MD rule: $i_k = \underset{1\le i\le m}{\operatorname{argmax}}\ d_k^{(i)}$, which gives the KMD method.

Geometrically speaking, the i-th entry $d_k^{(i)}$ of the distance vector d_k is the distance of the iteration point x_k from the i-th hyperplane described by the equation $A^{(i)}x = b^{(i)}$. Note that in the KMR and KMD methods we only choose the first index attaining the maximum in case that the maximum is attained at more than one point.

Assume that the number of nonzeros in the i-th row of the matrix $A \in \mathbb{C}^{m\times n}$ is nnz_i, and denote by $\mathrm{nnz} = \sum_{i=1}^{m} \mathrm{nnz}_i$. In Table 3, we list the computational costs in *floating point operations* (**flops**) for the k-th iterate of the Kaczmarz-type methods. Here, the maximum operation represents the operation that finds the first index of the maximal entry for a column vector $q \in \mathbb{C}^m$, with the corresponding computational cost being denoted by χ_m. As for the initializations of the Kaczmarz-type methods, we are required to compute the squared row norms $\|A^{(i)}\|^2$, $i = 1, 2, \ldots, m$, which are indeed not necessary to be repeatedly computed at each iteration step in the experiments.

3. The convergence properties

In this section, we restate the existing best upper bounds for the RKU, RKNU, RKR, RKD, KMR, and KMD methods with respect to the contraction and convergence factors of the mean-squared errors. Meanwhile, we also derive and provide several sharper upper bounds on the contraction and convergence rates for some of these Kaczmarz-type methods.

According to the convergence rates of the Kaczmarz-type methods, Strohmer and Vershynin [36] gave an upper bound for the RKNU method, Guo and Li [18] demonstrated that the RKR and RKNU methods can share the same upper bound. We refer to [19] for further improvement about the upper bound with respect to the RKR method, and [32] for a summary about the upper bounds with respect to the RKU, KMR, and KMD methods.

In order to restate these known results and also derive more accurate as well as new upper bounds for the convergence rates of the KMR, KMD, and RKD methods, we introduce the following necessary notation.

With respect to the matrix $A \in \mathbb{C}^{m\times n}$, we denote by

$$D = \mathrm{diag}(\|A^{(1)}\|, \|A^{(2)}\|, \ldots, \|A^{(m)}\|)$$

and

$$\tilde{A} = D^{-1}A.$$

Clearly, $D \in \mathbb{R}^{m\times m}$ is a diagonal matrix with its i-th diagonal entry being equal to the Euclidean norm of the i-th row of the matrix $A \in \mathbb{C}^{m\times n}$, and $\tilde{A} \in \mathbb{C}^{m\times n}$ is a left row-scaled matrix from the matrix $A \in \mathbb{C}^{m\times n}$ such that the Euclidean norms of all rows of the matrix $\tilde{A} \in \mathbb{C}^{m\times n}$ are equal to 1. Besides, let

$$\mathcal{H}_0(A) = \{u \mid u \neq 0 \text{ and } u \in \mathcal{R}(A^*)\},$$

and

$$\mathcal{H}(A) = \{u \mid u \in \mathcal{H}_0(A), \text{ and there exists an } i \in \{1, 2, \ldots, m\} \text{ such that } A^{(i)}u = 0\}. \tag{3.1}$$

Then we define the quantities

$$\sigma_{\mathcal{H}_0}(A) \triangleq \inf_{u \in \mathcal{H}_0(A)} \frac{\|Au\|_\infty}{\|u\|} = \inf_{u \in \mathcal{H}_0(A)} \frac{\max_i |A^{(i)}u|}{\|u\|}$$

and

$$\sigma_{\mathcal{H}}(A) \triangleq \inf_{u \in \mathcal{H}(A)} \frac{\|Au\|_\infty}{\|u\|} = \inf_{u \in \mathcal{H}(A)} \frac{\max_i |A^{(i)}u|}{\|u\|}. \tag{3.2}$$

We see that both $\sigma_{\mathcal{H}_0}(A)$ and $\sigma_{\mathcal{H}}(A)$ are restricted infinity norms of the matrix A with the restriction domains being $\mathcal{H}_0(A)$ and $\mathcal{H}(A)$, respectively. Note that

$$\mathcal{H}(A) = \mathcal{H}(\tilde{A}), \quad \mathcal{H}_0(A) = \mathcal{H}_0(\tilde{A}),$$

and it follows from $\mathcal{H}(A) \subseteq \mathcal{H}_0(A)$ that $\sigma_{\mathcal{H}_0}(A) \le \sigma_{\mathcal{H}}(A)$. In the subsequent discussions, we may simply write $\mathcal{H}(A)$ and $\mathcal{H}_0(A)$ as $\mathcal{H}$ and $\mathcal{H}_0$, if there is no confusion without explicitly showing up the matrix $A \in \mathbb{C}^{m\times n}$.

We define the *mean-squared distance* (**MSD**) between two adjacent iterates x_k and x_{k+1} by

$$\mathrm{MSD}_k \triangleq \mathbb{E}\|x_{k+1} - x_k\|^2,$$

and the *mean-squared error* (**MSE**) between the iterate x_k and the true solution x_* of the linear system (1.1) by

$$\mathrm{MSE}_k \triangleq \mathbb{E}\|x_* - x_k\|^2.$$

Here the true solution x_* means actually the least-norm solution of the consistent linear system (1.1), which can be explicitly expressed as $x_* = A^\dagger b$, with $A^\dagger$ indicating the Moore-Penrose pseudoinverse of the coefficient matrix $A \in \mathbb{C}^{m\times n}$. We remark that when $\{x_k\}_{k=0}^\infty$ is a non-randomized iteration sequence, MSD_k and MSE_k reduce to

$$\mathrm{MSD}_k \triangleq \|x_{k+1} - x_k\|^2 \quad \text{and} \quad \mathrm{MSE}_k \triangleq \|x_* - x_k\|^2,$$

the distance and the error, respectively. It is deserved to point out that for a fixed iterate x_k, the mean-squared error MSE_{k+1} is decreasing with respect to the increase of the mean-squared distance MSD_k; see the equality in (3.3).

The following lemma provides an important tool for deriving upper bounds about the MSE_k of the randomized Kaczmarz iteration sequences.

Lemma 3.1. *Let m be a prescribed positive integer, $\{\alpha_i\}_{i=1}^m$ be nonnegative reals, and $\{\beta_i\}_{i=1}^m$ be positive reals. Then the following inequalities hold true:*

(i) $\displaystyle\sum_{i=1}^m \frac{\alpha_i^2}{\beta_i} \ge \frac{\left(\sum_{i=1}^m \alpha_i\right)^2}{\sum_{i=1}^m \beta_i}$,

(ii) $\displaystyle\sum_{i=1}^m \alpha_i^2 \ge \frac{1}{m}\left(\sum_{i=1}^m \alpha_i\right)^2$.

In fact, the inequalities in Lemma 3.1 can be derived directly by making use of the Cauchy-Schwarz inequality [2], with the second one being particularly obtained by applying the Cauchy-Schwarz inequality to the nonnegative vectors $(1, 1, \ldots, 1)^T \in \mathbb{R}^m$ and $(\alpha_1, \alpha_2, \ldots, \alpha_m)^T \in \mathbb{R}^m$.

We now demonstrate upper bounds about the MSE_k for the six Kaczmarz-type methods, which are precisely stated in the following theorem.

Theorem 3.1. *Let $x_* \in \mathbb{C}^n$ be the least-norm solution of the consistent linear system (1.1), and $x_0 \in \mathcal{R}(A^*)$ be an initial guess. Then, at iteration step k ($k \ge 1$), the mean-squared errors of the Kaczmarz-type methods can be bounded as follows:*

(1) *for the RKU method [32],*

$$\mathbb{E}\|x_k - x_*\|^2 \le \left(1 - \frac{\sigma_{\min}^2(\tilde{A})}{m}\right)^k \|x_0 - x_*\|^2;$$

(2) *for the RKNU method [36],*

$$\mathbb{E}\|x_k - x_*\|^2 \le \left(1 - \frac{\sigma_{\min}^2(A)}{\|A\|_F^2}\right)^k \|x_0 - x_*\|^2;$$

(3) *for the RKR method [19],*

$$\mathbb{E}\|x_k - x_*\|^2 \le \left(1 - \frac{\sigma_{\min}^2(A)}{\gamma}\right)^{k-1} \left(1 - \frac{\sigma_{\min}^2(A)}{\|A\|_F^2}\right) \|x_0 - x_*\|^2,$$

with $\gamma = \|A\|_F^2 - \min\limits_{1 \le i \le m} \|A^{(i)}\|^2$;

(4) *for the RKD method,*

$$\mathbb{E}\|x_k - x_*\|^2 \le \left(1 - \frac{\sigma_{\min}^2(\tilde{A})}{m-1}\right)^{k-1} \left(1 - \frac{\sigma_{\min}^2(\tilde{A})}{m}\right) \|x_0 - x_*\|^2;$$

(5) *for the KMR method,*

$$\mathbb{E}\|x_k - x_*\|^2 \le \left(1 - \frac{\sigma_{\mathcal{H}}^2(A)}{\delta}\right)^{k-1} \left(1 - \frac{\sigma_{\mathcal{H}_0}^2(A)}{\delta}\right) \|x_0 - x_*\|^2,$$

with $\delta = \max\limits_{1 \le i \le m} \|A^{(i)}\|^2$;

(6) *for the KMD method,*

$$\mathbb{E}\|x_k - x_*\|^2 \le \left(1 - \sigma_{\mathcal{H}}^2(\tilde{A})\right)^{k-1} \left(1 - \sigma_{\mathcal{H}_0}^2(\tilde{A})\right) \|x_0 - x_*\|^2.$$

Proof. From the update formula of the RK method, we know that the increment vector $x_{k+1} - x_k$ is parallel to the i_k-th column $\left(A^{(i_k)}\right)^*$ of the matrix $A^* \in \mathbb{C}^{n \times m}$, and the error vector $x_{k+1} - x_*$ satisfies

$$A^{(i_k)}(x_{k+1} - x_*) = b^{(i_k)} - b^{(i_k)} = 0.$$

Note that the error vector $x_{k+1} - x_*$ is orthogonal to the i_k-th column $\left(A^{(i_k)}\right)^*$ of the matrix $A^* \in \mathbb{C}^{n \times m}$. Therefore, the two vectors $x_{k+1} - x_k$ and $x_{k+1} - x_*$ are mutually orthogonal, and in light of the Pythagorean theorem we have

$$\|x_{k+1} - x_*\|^2 = \|x_k - x_*\|^2 - \|x_{k+1} - x_k\|^2. \tag{3.3}$$

Recall that

$$\mathrm{MSD}_k = \|x_{k+1} - x_k\|^2 = \frac{|A^{(i_k)}(x_k - x_*)|^2}{\|A^{(i_k)}\|^2} = |\tilde{A}^{(i_k)}(x_k - x_*)|^2 \tag{3.4}$$

if $\{x_k\}_{k=0}^{\infty}$ is a non-randomized iteration sequence, and

$$\mathrm{MSD}_k = \mathbb{E}\|x_{k+1} - x_k\|^2 = \sum_{i_k=1}^{m} p_{i_k} \cdot \frac{|A^{(i_k)}(x_k - x_*)|^2}{\|A^{(i_k)}\|^2} = \sum_{i_k=1}^{m} p_{i_k} \cdot |\tilde{A}^{(i_k)}(x_k - x_*)|^2 \tag{3.5}$$

if $\{x_k\}_{k=0}^{\infty}$ is a randomized iteration sequence.

We now analyze MSD_k and derive the corresponding lower bounds with respect to the six Kaczmarz-type methods RKU, RKNU, RKR, RKD, KMR, and KMD, which are indicated as $\mathrm{MSD}_k^{\mathrm{RKU}}$, $\mathrm{MSD}_k^{\mathrm{RKNU}}$, $\mathrm{MSD}_k^{\mathrm{RKR}}$, $\mathrm{MSD}_k^{\mathrm{RKD}}$, $\mathrm{MSD}_k^{\mathrm{KMR}}$, and $\mathrm{MSD}_k^{\mathrm{KMD}}$. These lower bounds then straightforwardly lead to the upper bounds for MSE_k with respect to these six Kaczmarz-type methods, which are represented as $\mathrm{MSE}_k^{\mathrm{RKU}}$, $\mathrm{MSE}_k^{\mathrm{RKNU}}$, $\mathrm{MSE}_k^{\mathrm{RKR}}$, $\mathrm{MSE}_k^{\mathrm{RKD}}$, $\mathrm{MSE}_k^{\mathrm{KMR}}$, and $\mathrm{MSE}_k^{\mathrm{KMD}}$.

(1) For the RKU method, $p_{i_k} = \frac{1}{m}$, $i_k = 1, 2, \ldots, m$, so we have

$$\begin{aligned}\mathrm{MSD}_k^{\mathrm{RKU}} \triangleq \mathrm{MSD}_k &= \sum_{i_k=1}^{m} \frac{1}{m} \cdot |\tilde{A}^{(i_k)}(x_k - x_*)|^2 = \frac{1}{m}\|\tilde{A}(x_k - x_*)\|^2 \\ &\ge \frac{\sigma_{\min}^2(\tilde{A})}{m} \|x_k - x_*\|^2.\end{aligned} \tag{3.6}$$

(2) For the RKNU method, $p_{i_k} = \frac{\|A^{(i_k)}\|^2}{\|A\|_F^2}$, $i_k = 1, 2, \dots, m$, so we have

$$\begin{aligned}\mathrm{MSD}_k^{\mathrm{RKNU}} &\triangleq \mathrm{MSD}_k = \sum_{i_k=1}^{m} \frac{\|A^{(i_k)}\|^2}{\|A\|_F^2} \cdot \frac{|A^{(i_k)}(x_k - x_*)|^2}{\|A^{(i_k)}\|^2} \\ &= \frac{1}{\|A\|_F^2} \sum_{i_k=1}^{m} |A^{(i_k)}(x_k - x_*)|^2 = \frac{\|A(x_k - x_*)\|^2}{\|A\|_F^2} \\ &\geq \frac{\sigma_{\min}^2(A)}{\|A\|_F^2} \|x_k - x_*\|^2. \end{aligned} \tag{3.7}$$

(3) For the RKR method, $p_{i_k} = \frac{|r_k^{(i_k)}|^2}{\|r_k\|^2}$, $i_k = 1, 2, \dots, m$, so from the equalities $r_k = A(x_* - x_k)$ and $r_k^{(i_{k-1})} = 0$ we have, for $k \geq 1$, that

$$\mathrm{MSD}_k^{\mathrm{RKR}} \triangleq \mathrm{MSD}_k = \sum_{i_k=1}^{m} \frac{|r_k^{(i_k)}|^2}{\|r_k\|^2} \cdot \frac{|A^{(i_k)}(x_k - x_*)|^2}{\|A^{(i_k)}\|^2} = \frac{1}{\|r_k\|^2} \sum_{\substack{i=1 \\ i \neq i_{k-1}}}^{m} \frac{|r_k^{(i)}|^4}{\|A^{(i)}\|^2}. \tag{3.8}$$

With application of Lemma 3.1 (i), we know that

$$\sum_{\substack{i=1 \\ i \neq i_{k-1}}}^{m} \|A^{(i)}\|^2 \cdot \sum_{\substack{i=1 \\ i \neq i_{k-1}}}^{m} \frac{|r_k^{(i)}|^4}{\|A^{(i)}\|^2} \geq \left(\sum_{\substack{i=1 \\ i \neq i_{k-1}}}^{m} \|A^{(i)}\| \cdot \frac{|r_k^{(i)}|^2}{\|A^{(i)}\|} \right)^2 = \left(\sum_{\substack{i=1 \\ i \neq i_{k-1}}}^{m} |r_k^{(i)}|^2 \right)^2 = \|r_k\|^4.$$

Here the last equality is valid due to the identity $r_k^{(i_{k-1})} = 0$. It then follows that

$$\frac{1}{\|r_k\|^2} \sum_{\substack{i=1 \\ i \neq i_{k-1}}}^{m} \frac{|r_k^{(i)}|^4}{\|A^{(i)}\|^2} \geq \frac{1}{\|r_k\|^2} \cdot \frac{\|r_k\|^4}{\sum\limits_{\substack{i=1 \\ i \neq i_{k-1}}}^{m} \|A^{(i)}\|^2} = \frac{\|r_k\|^2}{\sum\limits_{\substack{i=1 \\ i \neq i_{k-1}}}^{m} \|A^{(i)}\|^2}.$$

After substitution of this inequality into (3.8), we can further get the estimate

$$\mathrm{MSD}_k^{\mathrm{RKR}} \geq \frac{\|r_k\|^2}{\sum\limits_{\substack{i=1 \\ i \neq i_{k-1}}}^{m} \|A^{(i)}\|^2} \geq \frac{\|r_k\|^2}{\gamma} = \frac{\|A(x_* - x_k)\|^2}{\gamma} \geq \frac{\sigma_{\min}^2(A)}{\gamma} \|x_k - x_*\|^2. \tag{3.9}$$

In particular, for $k = 0$, in an analogous fashion we have

$$\begin{aligned}\mathrm{MSD}_0^{\mathrm{RKR}} &\triangleq \mathrm{MSD}_0 = \sum_{i_0=1}^{m} \frac{|r_0^{(i_0)}|^2}{\|r_0\|^2} \cdot \frac{|A^{(i_0)}(x_0 - x_*)|^2}{\|A^{(i_0)}\|^2} = \frac{1}{\|r_0\|^2} \sum_{i=1}^{m} \frac{|r_0^{(i)}|^4}{\|A^{(i)}\|^2} \\ &\geq \frac{\|r_0\|^2}{\|A\|_F^2} = \frac{\|A(x_* - x_0)\|^2}{\|A\|_F^2} \geq \frac{\sigma_{\min}^2(A)}{\|A\|_F^2} \|x_0 - x_*\|^2. \end{aligned} \tag{3.10}$$

Here we have made use of Lemma 3.1 (i) again in the derivation of the first inequality of (3.10).

(4) For the RKD method, $p_{i_k} = \frac{\left(d_k^{(i_k)}\right)^2}{\|d_k\|^2}$, $i_k = 1, 2, \dots, m$, so from $d_k^{(i_k)} = |\tilde{A}^{(i_k)}(x_k - x_*)|$ and $d_k^{(i_{k-1})} = 0$ we have

$$\begin{aligned}\mathrm{MSD}_k^{\mathrm{RKD}} &\triangleq \mathrm{MSD}_k = \sum_{i_k=1}^{m} \frac{\left(d_k^{(i_k)}\right)^2}{\|d_k\|^2} \cdot |\tilde{A}^{(i_k)}(x_k - x_*)|^2 = \frac{1}{\|d_k\|^2} \sum_{i=1}^{m} \left(d_k^{(i)}\right)^2 |\tilde{A}^{(i)}(x_k - x_*)|^2 \\ &= \frac{1}{\|d_k\|^2} \sum_{\substack{i=1 \\ i \neq i_{k-1}}}^{m} \left(d_k^{(i)}\right)^4 \geq \frac{1}{\|d_k\|^2} \cdot \frac{1}{m-1} \left(\sum_{\substack{i=1 \\ i \neq i_{k-1}}}^{m} \left(d_k^{(i)}\right)^2 \right)^2 \\ &= \frac{1}{m-1} \|d_k\|^2 = \frac{1}{m-1} \|\tilde{A}(x_k - x_*)\|^2 \geq \frac{\sigma_{\min}^2(\tilde{A})}{m-1} \|x_k - x_*\|^2 \end{aligned} \tag{3.11}$$

for $k \geq 1$, and

$$\mathrm{MSD}_0^{\mathrm{RKD}} \triangleq \mathrm{MSD}_0 = \sum_{i_0=1}^{m} \frac{\left(d_0^{(i_0)}\right)^2}{\|d_0\|^2} \cdot |\tilde{A}^{(i_0)}(x_0 - x_*)|^2 = \frac{\sum\limits_{i=1}^{m}\left(d_0^{(i)}\right)^4}{\sum\limits_{i=1}^{m}\left(d_0^{(i)}\right)^2}$$
$$\geq \frac{1}{m}\sum_{i=1}^{m}\left(d_0^{(i)}\right)^2 = \frac{1}{m}\|\tilde{A}(x_0 - x_*)\|^2 \geq \frac{\sigma_{\min}^2(\tilde{A})}{m}\|x_0 - x_*\|^2. \tag{3.12}$$

Here, in the derivations of the first inequalities in (3.11) and (3.12), we have made use of Lemma 3.1 (ii).

(5) For the KMR method, $i_k = \operatorname*{argmax}_{1\leq i\leq m} |r_k^{(i)}|$, so we have

$$\mathrm{MSD}_k^{\mathrm{KMR}} \triangleq \mathrm{MSD}_k = \frac{\max\limits_{1\leq i\leq m}|r_k^{(i)}|^2}{\|A^{(i_k)}\|^2} = \frac{\|A(x_* - x_k)\|_\infty^2}{\|A^{(i_k)}\|^2}$$
$$\geq \frac{\|A(x_k - x_*)\|_\infty^2}{\delta} \geq \frac{\sigma_{\mathcal{H}}^2(A)}{\delta}\|x_k - x_*\|^2 \tag{3.13}$$

for $k \geq 1$. Here the last inequality holds true due to the fact that $x_k - x_* \in \mathcal{H}(A)$. For $k = 0$, as $x_0 - x_* \in \mathcal{H}_0(A)$, we have

$$\mathrm{MSD}_0^{\mathrm{KMR}} \triangleq \mathrm{MSD}_0 = \frac{\max\limits_{1\leq i\leq m}|r_0^{(i)}|^2}{\|A^{(i_0)}\|^2} = \frac{\|A(x_* - x_0)\|_\infty^2}{\|A^{(i_0)}\|^2}$$
$$\geq \frac{\|A(x_0 - x_*)\|_\infty^2}{\delta} \geq \frac{\sigma_{\mathcal{H}_0}^2(A)}{\delta}\|x_0 - x_*\|^2. \tag{3.14}$$

(6) For the KMD method, $i_k = \operatorname*{argmax}_{1\leq i\leq m} d_k^{(i)}$, so from $x_k - x_* \in \mathcal{H}(\tilde{A})$ and $x_0 - x_* \in \mathcal{H}_0(\tilde{A})$ we have

$$\mathrm{MSD}_k^{\mathrm{KMD}} \triangleq \mathrm{MSD}_k = \max_{1\leq i\leq m}\left(d_k^{(i)}\right)^2 = \|d_k\|_\infty^2$$
$$= \|\tilde{A}(x_k - x_*)\|_\infty^2 \geq \sigma_{\mathcal{H}}^2(\tilde{A})\,\|x_k - x_*\|^2 \tag{3.15}$$

for $k \geq 1$, and

$$\mathrm{MSD}_0^{\mathrm{KMD}} \triangleq \mathrm{MSD}_0 = \max_{1\leq i\leq m}\left(d_0^{(i)}\right)^2 = \|d_0\|_\infty^2$$
$$= \|\tilde{A}(x_0 - x_*)\|_\infty^2 \geq \sigma_{\mathcal{H}_0}^2(\tilde{A})\,\|x_0 - x_*\|^2. \tag{3.16}$$

Up to now, with substitutions of the estimates in (3.6)-(3.16) into (3.3), by recursively operating with respect to the iterate index k, we can obtain the upper bounds in (1)-(6) of Theorem 3.1. □

Remark 3.1. The estimates for the convergence rates of the RKU, RKNU, and RKR methods given in Theorem 3.1 are the same as those in [32], [36], and [19], respectively. However, by introducing the concept MSD_k we here give unified proofs for all of these Kaczmarz-type methods. Moreover, the estimate for the convergence rate of the RKD method is new and given first in this work, and the estimates for the convergence rates of the KMR and KMD methods given in Theorem 3.1 are more precise than the existing ones given in [32].

We define the *mean-squared root-convergence factor* of the Kaczmarz-type methods by

$$\mathrm{MRF} \triangleq \lim_{k\to\infty}\left(\frac{\mathbb{E}\|x_k - x_*\|^2}{\|x_0 - x_*\|^2}\right)^{1/k} = \lim_{k\to\infty}\left(\frac{\mathrm{MSE}_k}{\mathrm{MSE}_0}\right)^{1/k},$$

and denote its upper bound as BMRF. We remark that for the non-randomized Kaczmarz-type methods, the corresponding MRF is indeed the squared root-convergence factor, and is still indicated and defined as

$$\mathrm{MRF} \triangleq \lim_{k\to\infty}\left(\frac{\|x_k - x_*\|^2}{\|x_0 - x_*\|^2}\right)^{1/k}.$$

Then, corresponding to the upper bounds of MSE_k for the RKU, RKNU, RKR, RKD, KMR, and KMD methods, that is, $\mathrm{MSE}_k^{\mathrm{RKU}}$, $\mathrm{MSE}_k^{\mathrm{RKNU}}$, $\mathrm{MSE}_k^{\mathrm{RKR}}$, $\mathrm{MSE}_k^{\mathrm{RKD}}$, $\mathrm{MSE}_k^{\mathrm{KMR}}$, and $\mathrm{MSE}_k^{\mathrm{KMD}}$ given in Theorem 3.1, we can obtain upper bounds $\mathrm{BMRF}^{\mathrm{RKU}}$, $\mathrm{BMRF}^{\mathrm{RKNU}}$, $\mathrm{BMRF}^{\mathrm{RKR}}$, $\mathrm{BMRF}^{\mathrm{RKD}}$, $\mathrm{BMRF}^{\mathrm{KMR}}$, and $\mathrm{BMRF}^{\mathrm{KMD}}$ for the mean-squared root-convergence rates of the Kaczmarz-type methods RKU, RKNU, RKR, RKD, KMR, and KMD, which are specifically given as follows:

$$\begin{cases} \mathrm{BMRF}^{\mathrm{RKU}} = 1 - \frac{\sigma_{\min}^2(\tilde{A})}{m}, \ \mathrm{BMRF}^{\mathrm{RKNU}} = 1 - \frac{\sigma_{\min}^2(A)}{\|A\|_F^2}, \ \mathrm{BMRF}^{\mathrm{RKR}} = 1 - \frac{\sigma_{\min}^2(A)}{\gamma}, \\ \mathrm{BMRF}^{\mathrm{RKD}} = 1 - \frac{\sigma_{\min}^2(\tilde{A})}{m-1}, \ \mathrm{BMRF}^{\mathrm{KMR}} = 1 - \frac{\sigma_{\mathcal{H}}^2(A)}{\delta}, \quad \mathrm{BMRF}^{\mathrm{KMD}} = 1 - \sigma_{\mathcal{H}}^2(\tilde{A}). \end{cases} \tag{3.17}$$

Remark 3.2. In fact, the upper bounds for the convergence rates of the KMR and KMD methods given by Nutini, Sepehry, and Laradji, et al., in [32] are as follows:

$$\mathrm{BMRF}_o^{\mathrm{KMR}} = 1 - \frac{\sigma(A,\infty)^2}{\max\limits_{1\le i\le m} \|A^{(i)}\|^2} \quad \text{and} \quad \mathrm{BMRF}_o^{\mathrm{KMD}} = 1 - \sigma(\tilde{A},\infty)^2,$$

where

$$\sigma(A,\infty) \triangleq \inf_{x\notin \mathbb{S}} \frac{\|A(x-x_*)\|_\infty}{\|x-x_*\|},$$

with $\sigma(\tilde{A},\infty)$ being defined similarly. Here $\mathbb{S}$ and x_* are, respectively, the solution space and the least-norm solution of the consistent linear system $Ax=b$. Note that if $x-x_*\in\mathcal{R}(A^*)$ and $x\neq x_*$, i.e., $x-x_*\in\mathcal{H}_0(A)$, then it holds that

$$\|b-Ax\| = \|A(x-x_*)\| \ge \sigma_{\min}(A)\,\|x-x_*\| > 0,$$

which infers that $b-Ax\neq 0$, i.e., $x\notin\mathbb{S}$. Combining this notice with the fact that $\mathcal{H}(A)\subseteq\mathcal{H}_0(A)$, we have

$$\sigma_{\mathcal{H}}(A) \ge \sigma_{\mathcal{H}_0}(A) \ge \sigma(A,\infty),$$

which further indicates that $\mathrm{BMRF}^{\mathrm{KMR}} \le \mathrm{BMRF}_o^{\mathrm{KMR}}$. Similarly, we can obtain $\mathrm{BMRF}^{\mathrm{KMD}} \le \mathrm{BMRF}_o^{\mathrm{KMD}}$.

As a matter of fact, at each iteration step k, we expect that the next iterate x_{k+1} can be as close as possible to the exact solution x_*. In other words, we expect that MSD_k can be closer to its maximum at least at the early stage of the iteration process, while MSE_{k+1} can be closer to its minimum. With respect to this regard, we see that the optimal choice could be provided by the MD selection rule. Therefore, it turns out that the KMD method should be the best among the six Kaczmarz-type methods with respect to the evaluation criterion for MSD_k.

The following theorem provides rigorous comparisons about the mean-squared distances and the mean-squared root-convergence factors for the Kaczmarz-type methods.

Theorem 3.2. *The following statements hold true:*

(i) $\mathrm{MSD}_k^{\mathrm{RKD}} > \mathrm{MSD}_k^{\mathrm{RKU}}$, $\mathrm{MSD}_k^{\mathrm{RKR}} > \mathrm{MSD}_k^{\mathrm{RKNU}}$, $k=1,2,\ldots$;
(ii) $\mathrm{BMRF}^{\mathrm{RKD}} < \mathrm{BMRF}^{\mathrm{RKU}}$, $\mathrm{BMRF}^{\mathrm{RKR}} < \mathrm{BMRF}^{\mathrm{RKNU}}$;
(iii) $\mathrm{BMRF}^{\mathrm{KMD}} \le \min\{\mathrm{BMRF}^{\mathrm{RKU}}, \mathrm{BMRF}^{\mathrm{RKNU}}, \mathrm{BMRF}^{\mathrm{RKR}}, \mathrm{BMRF}^{\mathrm{RKD}}, \mathrm{BMRF}^{\mathrm{KMR}}\}$.

Proof. It follows from (3.11) and (3.6) that

$$\mathrm{MSD}_k^{\mathrm{RKD}} \ge \frac{1}{m-1}\|\tilde{A}(x_k-x_*)\|^2 > \frac{1}{m}\|\tilde{A}(x_k-x_*)\|^2 = \mathrm{MSD}_k^{\mathrm{RKU}}.$$

Since the coefficient matrix $A\in\mathbb{C}^{m\times n}$ does not have any zero row, we have

$$\gamma = \|A\|_F^2 - \min_{1\le i\le m}\|A^{(i)}\|^2 < \|A\|_F^2. \tag{3.18}$$

Hence, it follows from (3.9) and (3.7) that

$$\mathrm{MSD}_k^{\mathrm{RKR}} \ge \frac{1}{\gamma}\|A(x_*-x_k)\|^2 > \frac{\|A(x_k-x_*)\|^2}{\|A\|_F^2} = \mathrm{MSD}_k^{\mathrm{RKNU}}.$$

This demonstrates the validity of Statement (i).

The relationships in Statement (ii) follow straightforwardly from (3.17), as it holds that

$$\mathrm{BMRF}^{\mathrm{RKD}} = 1 - \frac{\sigma_{\min}^2(\tilde{A})}{m-1} < 1 - \frac{\sigma_{\min}^2(\tilde{A})}{m} = \mathrm{BMRF}^{\mathrm{RKU}}$$

and

$$\mathrm{BMRF}^{\mathrm{RKR}} = 1 - \frac{\sigma_{\min}^2(A)}{\gamma} < 1 - \frac{\sigma_{\min}^2(A)}{\|A\|_F^2} = \mathrm{BMRF}^{\mathrm{RKNU}}.$$

We now turn to prove Statement (iii). To this end, for any $u \in \mathcal{H}$ we assume that there exists a row index $i_u \in \{1, 2, \ldots, m\}$ of the coefficient matrix $A \in \mathbb{C}^{m\times n}$ such that $A^{(i_u)}u = 0$. Then, for any positive constant ν, it holds that

$$\frac{\|Au\|^2}{\nu} = \sum_{\substack{i=1\\ i\neq i_u}}^{m} \frac{|A^{(i)}u|^2}{\|A^{(i)}\|^2} \cdot \frac{\|A^{(i)}\|^2}{\nu} \le \frac{\|A\|_F^2 - \|A^{(i_u)}\|^2}{\nu} \cdot \max_{1\le i\le m} \frac{|A^{(i)}u|^2}{\|A^{(i)}\|^2} \le \frac{\gamma}{\nu} \cdot \max_{1\le i\le m} |\tilde{A}^{(i)}u|^2.$$

By letting $\nu = \gamma$, according to the definition of $\sigma_{\mathcal{H}}(\tilde{A})$ we then have the estimate

$$\sigma_{\mathcal{H}}^2(\tilde{A}) = \inf_{u\in\mathcal{H}} \frac{\max\limits_{1\le i\le m} |\tilde{A}^{(i)}u|^2}{\|u\|^2} \ge \frac{1}{\gamma} \cdot \inf_{u\in\mathcal{H}} \frac{\|Au\|^2}{\|u\|^2} \ge \frac{1}{\gamma} \cdot \sigma_{\min}^2(A). \tag{3.19}$$

This immediately implies that $\mathrm{BMRF}^{\mathrm{KMD}} \le \mathrm{BMRF}^{\mathrm{RKR}}$. Here we point out that $\sigma_{\min}(A)$ is defined by

$$\sigma_{\min}(A) \triangleq \inf_{u\in\mathcal{H}_0} \frac{\|Au\|}{\|u\|}$$

and $\mathcal{H}$ is a subspace of $\mathcal{H}_0$. Moreover, it follows from (3.18) and (3.19) that

$$\sigma_{\mathcal{H}}^2(\tilde{A}) \ge \frac{1}{\gamma} \cdot \sigma_{\min}^2(A) \ge \frac{1}{\|A\|_F^2} \cdot \sigma_{\min}^2(A).$$

So, we can obtain the relationship $\mathrm{BMRF}^{\mathrm{KMD}} \le \mathrm{BMRF}^{\mathrm{RKNU}}$.

Based upon the definition of $\sigma_{\mathcal{H}}(\tilde{A})$ and the inequality

$$\|\tilde{A}u\|_\infty^2 \ge \frac{1}{m-1}\|\tilde{A}u\|^2, \quad \forall u \in \mathcal{H},$$

we have

$$\sigma_{\mathcal{H}}^2(\tilde{A}) \ge \frac{1}{m-1} \cdot \inf_{u\in\mathcal{H}} \frac{\|\tilde{A}u\|^2}{\|u\|^2} \ge \frac{1}{m-1} \cdot \sigma_{\min}^2(\tilde{A}),$$

which immediately implies $\mathrm{BMRF}^{\mathrm{KMD}} \le \mathrm{BMRF}^{\mathrm{RKD}}$.

The validity of the relationship $\mathrm{BMRF}^{\mathrm{KMD}} \le \mathrm{BMRF}^{\mathrm{RKU}}$ is obvious due to the inequality

$$\sigma_{\mathcal{H}}^2(\tilde{A}) \ge \frac{1}{m} \cdot \sigma_{\min}^2(\tilde{A}), \quad \text{for any} \quad m \ge 1.$$

Finally, the validity of the relationship $\mathrm{BMRF}^{\mathrm{KMD}} \le \mathrm{BMRF}^{\mathrm{KMR}}$ follows directly from the estimate

$$\max_{1\le i\le m} |\tilde{A}^{(i)}u|^2 = \max_{1\le i\le m} \frac{|A^{(i)}u|^2}{\|A^{(i)}\|^2} \ge \frac{\max\limits_{1\le i\le m} |A^{(i)}u|^2}{\delta},$$

or in other words,

$$\sigma_{\mathcal{H}}^2(\tilde{A}) \ge \frac{\sigma_{\mathcal{H}}^2(A)}{\delta}. \quad \square$$

From Theorem 3.2, we can learn that the R rule and the D rule are better than the NU rule and the U rule, respectively, while the MD rule is the best among all of these six selection rules, with respect to both MSD_k and BMRF.

Remark 3.3. If all rows of the coefficient matrix $A \in \mathbb{C}^{m\times n}$ have the same norms, then the NU, D, and MD rules automatically degenerate to the U, R, and MR rules, respectively. Furthermore, it holds that

(i) $\mathrm{MSD}_k^{\mathrm{KMD}} = \mathrm{MSD}_k^{\mathrm{KMR}} \ge \mathrm{MSD}_k^{\mathrm{RKD}} = \mathrm{MSD}_k^{\mathrm{RKR}} \ge \mathrm{MSD}_k^{\mathrm{RKNU}} = \mathrm{MSD}_k^{\mathrm{RKU}}$,
(ii) $\mathrm{BMRF}^{\mathrm{KMD}} = \mathrm{BMRF}^{\mathrm{KMR}} \le \mathrm{BMRF}^{\mathrm{RKD}} = \mathrm{BMRF}^{\mathrm{RKR}} < \mathrm{BMRF}^{\mathrm{RKNU}} = \mathrm{BMRF}^{\mathrm{RKU}}$.

In particular, for $k \ge 1$, we have $\mathrm{MSD}_k^{\mathrm{RKR}} > \mathrm{MSD}_k^{\mathrm{RKNU}}$, that is, the last inequality in (i) becomes strict.

4. Comparisons with specific examples

In this section, with specific examples we explain that the comparable relationships for the U, NU, R, D, and MR rules are problem-dependent in terms of both evaluation criterions MSD and BMRF, except for the quantitatively determined relationships shown in Section 3. To this end, we consider a family of real and square matrices $A \in \mathbb{R}^{m\times m}$ of the form

$$A = \begin{bmatrix} \alpha+1 & 1 & \cdots & 1 \\ 1 & \beta+1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & \beta+1 \end{bmatrix} = \mathrm{diag}(\alpha, \beta, \dots, \beta) + \mathbf{1}_m \mathbf{1}_m^T \triangleq H_m + \mathbf{1}_m \mathbf{1}_m^T, \tag{4.1}$$

where α and β are arbitrary reals, $H_m = \mathrm{diag}(\alpha, \beta, \dots, \beta) \in \mathbb{R}^{m\times m}$ is a diagonal matrix with its first diagonal entry being equal to α and the other diagonal entries being all equal to β, and $\mathbf{1}_m$ is the m-dimensional column vector of all entries being equal to 1. Note that the diagonal entries of the matrix $A \in \mathbb{R}^{m\times m}$ are the same from the second to the last entry, and the off-diagonal entries of the matrix $A \in \mathbb{R}^{m\times m}$ are all equal to 1. For convenience of our expression, in the sequel we may neglect the subscript of the vector $\mathbf{1}_m$ if its dimension is clear from the context.

4.1. Comparable relationships on MSD_0

For a fixed iteration index k, it follows from (3.4) and (3.5) that the value of MSD_k depends on the iterate x_k, while the iterate x_k, in turn, depends on an actual selection rule adopted in a referred Kaczmarz-type method. In general, different selection rules may select different rows in computing the current iterate x_k; and even though some selection rules may choose the same working row such that the iterate x_k is the same, their follow-up performances are still dependent on their own selection rules of the working rows, leading possibly to different iterates. Hence, in the discussion of this subsection, we only focus on a one-step performance of certain selection rule and, for simplicity, we take MSD_0 as an example. Though this, our analyses and conclusions hold equally true for any fixed iteration index k.

Theorem 4.1. *For the matrix $A \in \mathbb{R}^{m\times m}$ as shown in (4.1), assume that $\alpha = -m$ and $\beta = -m$ are not satisfied simultaneously. Let the linear system (1.1) with this coefficient matrix have the exact solution $x_* = \mathbf{1}_m$. Set the right-hand side $b \in \mathbb{R}^m$ of the linear system (1.1) be $b = Ax_* = A\mathbf{1}_m$, and the initial guess $x_0 \in \mathbb{R}^m$ be zero. Denote by $t_\pm = -m \pm \sqrt{m(m-1)}$. Then the following statements hold true:*

(1) *if $\alpha, \beta \in (-\infty, t_-)$ or $\alpha, \beta \in (t_+, 0)$, we have*

$$\mathrm{MSD}_0^{\mathrm{KMR}} \geq \mathrm{MSD}_0^{\mathrm{RKR}} \geq \mathrm{MSD}_0^{\mathrm{RKNU}} \geq \mathrm{MSD}_0^{\mathrm{RKD}} \geq \mathrm{MSD}_0^{\mathrm{RKU}};$$

(2) *if $\alpha, \beta \in (t_-, -m)$ or $\alpha, \beta \in (-1, t_+)$, we have*

$$\mathrm{MSD}_0^{\mathrm{KMR}} \geq \mathrm{MSD}_0^{\mathrm{RKR}} \geq \mathrm{MSD}_0^{\mathrm{RKD}} \geq \mathrm{MSD}_0^{\mathrm{RKNU}} \geq \mathrm{MSD}_0^{\mathrm{RKU}};$$

(3) *if $\alpha, \beta \in (-m, -1)$, we have*

$$\mathrm{MSD}_0^{\mathrm{KMR}} \geq \mathrm{MSD}_0^{\mathrm{RKD}} \geq \mathrm{MSD}_0^{\mathrm{RKR}} \geq \mathrm{MSD}_0^{\mathrm{RKU}} \geq \mathrm{MSD}_0^{\mathrm{RKNU}};$$

(4) *if $\alpha, \beta \in (0, \infty)$, we have*

$$\mathrm{MSD}_0^{\mathrm{RKD}} \geq \mathrm{MSD}_0^{\mathrm{RKU}} \geq \mathrm{MSD}_0^{\mathrm{RKR}} \geq \mathrm{MSD}_0^{\mathrm{RKNU}} \geq \mathrm{MSD}_0^{\mathrm{KMR}}.$$

Proof. See Section A.1. □

In Table 4, we list the comparable relationships among the Kaczmarz-type methods in terms of MSD_0 as shown in Theorem 4.1. In this table, the symbol "$\leq$" or "$\geq$" without a superscript shows that the corresponding inequality relation is valid unconditionally, while the symbol "$\leq$" or "$\geq$" with one or more superscripts shows that the corresponding inequality relation is valid only conditionally. For example, "$\leq^{(1)}$" with the superscript '(1)' means that this inequality relation holds under the condition in Theorem 4.1 (1), and "$\leq^{(1),(2)}$" with the superscripts '(1), (2)' means that this inequality relation holds under the conditions in Theorem 4.1 (1) and (2). More specifically, from Table 4 we observe that $\mathrm{MSD}_0^{\mathrm{RKU}} \leq \mathrm{MSD}_0^{\mathrm{RKNU}}$ under the conditions in Theorem 4.1 (1) and (2), and $\mathrm{MSD}_0^{\mathrm{RKNU}} \leq \mathrm{MSD}_0^{\mathrm{RKU}}$ under the conditions in Theorem 4.1 (3) and (4). Evidently, there is no comparable relationship between $\mathrm{MSD}_0^{\mathrm{RKU}}$ and $\mathrm{MSD}_0^{\mathrm{RKNU}}$ in general, and only a problem-dependent relationship may be possible between them.

Indeed, from Table 4, we find that some comparable relationships among these Kaczmarz-type methods are problem-dependent. For an intuitive perspicuity, we list these comparable relationships in Table 5. In this table, we use the symbol "$\leq$" without a superscript to denote that the value of MSD_0 in that row is less than that in the corresponding column, or

Table 4
The comparable relationships in terms of MSD_0 as shown in Theorem 4.1.

Method	RKU	RKNU	RKR	RKD	KMR
RKU		$\le^{(1),(2)}$	$\le^{(1),(2),(3)}$	$\le$	$\le^{(1),(2),(3)}$
RKNU	$\le^{(3),(4)}$		$\le$	$\le^{(2),(3),(4)}$	$\le^{(1),(2),(3)}$
RKR	$\le^{(4)}$	$\ge$		$\le^{(3),(4)}$	$\le^{(1),(2),(3)}$
RKD	$\ge$	$\le^{(1)}$	$\le^{(1),(2)}$		$\le^{(1),(2),(3)}$
KMR	$\le^{(4)}$	$\le^{(4)}$	$\le^{(4)}$	$\le^{(4)}$	

Table 5
The skeleton for the comparable relationships in terms of MSD_0 as shown in Table 4.

Method	RKU	RKNU	RKR	RKD	KMR
RKU		–	–	$\le$	–
RKNU			$\le$	–	–
RKR				–	–
RKD					–
KMR					

in other words, the method in that column is better than the method in the corresponding row in terms of MSD_0. Besides, we use the symbol "–" to denote that which value of MSD_0 is smaller is problem-dependent. It is clear from Table 5 that the quantitative relationships among these five Kaczmarz-type methods in terms of MSD_0 are problem-dependent, except for the deterministic relationships $\mathrm{MSD}_0^{\mathrm{RKU}} \le \mathrm{MSD}_0^{\mathrm{RKD}}$ and $\mathrm{MSD}_0^{\mathrm{RKNU}} \le \mathrm{MSD}_0^{\mathrm{RKR}}$; see also Theorem 3.2.

Based on the skeleton intuitively shown in Table 5, we know that Theorem 4.1 implies that there is no deterministic relationships between any pairs of MSD_0 with respect to the RKU, RKNU, RKR, RKD, and KMR methods, except for the pairs $\mathrm{MSD}_0^{\mathrm{RKU}}$ and $\mathrm{MSD}_0^{\mathrm{RKD}}$, and $\mathrm{MSD}_0^{\mathrm{RKNU}}$ and $\mathrm{MSD}_0^{\mathrm{RKR}}$. Moreover, we observe that if the largest entry of the residual vector corresponds to a row whose norm is significantly large, then the MR rule may be the worst; see Statement (4) in Theorem 4.1. Otherwise, the MR rule can be the best among the five selection rules N, NU, R, D, and MR, and it is the closest to the MD rule in terms of MSD_0.

4.2. Comparable relationships on BMRF

As defined in (3.17), the upper bounds for the mean-squared root-convergence rates of the randomized Kaczmarz-type methods RKU, RKNU, RKR, and RKD, and the upper bounds for the root-convergence rates of the non-randomized Kaczmarz-type methods KMR and KMD, denoted as $\mathrm{BMRF}^{\mathrm{RKU}}$, $\mathrm{BMRF}^{\mathrm{RKNU}}$, $\mathrm{BMRF}^{\mathrm{RKR}}$, $\mathrm{BMRF}^{\mathrm{RKD}}$, $\mathrm{BMRF}^{\mathrm{KMR}}$, and $\mathrm{BMRF}^{\mathrm{KMD}}$, are independent of the iterate x_k, but are dependent on the actual selection rules of the working rows adopted in them. In fact, these bounds are completely determined by the algebraic properties about the coefficient matrix $A \in \mathbb{C}^{m\times n}$ of the linear system (1.1). In Theorem 3.2 (ii)-(iii), we have demonstrated the deterministic comparable relationships between $\mathrm{BMRF}^{\mathrm{RKU}}$ and $\mathrm{BMRF}^{\mathrm{RKD}}$, $\mathrm{BMRF}^{\mathrm{RKNU}}$ and $\mathrm{BMRF}^{\mathrm{RKR}}$, and $\mathrm{BMRF}^{\mathrm{KMD}}$ and the others. We now use the matrix $A \in \mathbb{R}^{m\times m}$ as described in (4.1) to further show that all of the other relationships are not deterministically comparable and, in fact, they are problem-dependent.

Lemma 4.1. *For the matrix $A \in \mathbb{R}^{m\times m}$ as shown in (4.1), let $\alpha > 0$ and $\beta = 0$. Denote by*

$$D \triangleq \mathrm{diag}\left(\|A^{(1)}\|, \|A^{(2)}\|, \ldots, \|A^{(m)}\|\right) = \mathrm{diag}\left(\sqrt{(\alpha+1)^2+m-1}, \sqrt{m}, \ldots, \sqrt{m}\right)$$

and

$$\tilde{A} \triangleq D^{-1}A = \frac{1}{\sqrt{m}}\begin{bmatrix} \frac{(\alpha+1)\sqrt{m}}{\sqrt{\alpha^2+2\alpha+m}} & \frac{\sqrt{m}}{\sqrt{\alpha^2+2\alpha+m}}\mathbf{1}_{m-1}^T \\ \mathbf{1}_{m-1} & \mathbf{1}_{m-1}\mathbf{1}_{m-1}^T \end{bmatrix}.$$

Then the following statements hold true:

(1) *the nonzero eigenvalues and the nonzero singular values of the matrix $A \in \mathbb{R}^{m\times m}$ are the same, which are*

$$\sigma_{\pm}(A) \equiv \lambda_{\pm}(A) = \frac{1}{2}\left(\alpha + m \pm \sqrt{(\alpha-m)^2+4\alpha}\right), \tag{4.2}$$

and the nonzero singular values of the matrix $\tilde{A} \in \mathbb{R}^{m\times m}$ are

$$\sigma_{\pm}(\tilde{A}) = \frac{\sqrt{2}}{2}\sqrt{m \pm \sqrt{m^2 - \frac{4(m-1)^2\alpha^2}{m(\alpha^2+2\alpha+m)}}};$$

(2) *the smallest nonzero singular values of the matrices* $A \in \mathbb{R}^{m\times m}$ *and* $\tilde{A} \in \mathbb{R}^{m\times m}$ *are*

$$\sigma_{\min}(A) = \lambda_-(A) = \frac{1}{2}\left(\alpha + m - \sqrt{(\alpha - m)^2 + 4\alpha}\right) \tag{4.3}$$

and

$$\sigma_{\min}(\tilde{A}) = \sigma_-(\tilde{A}) = \frac{\sqrt{2}}{2}\sqrt{m - \sqrt{m^2 - \frac{4(m-1)^2\alpha^2}{m(\alpha^2 + 2\alpha + m)}}}; \tag{4.4}$$

(3) *the restricted infinity norms of the matrices* $A \in \mathbb{R}^{m\times m}$ *and* $\tilde{A} \in \mathbb{R}^{m\times m}$ *are*

$$\sigma_{\mathcal{H}}(A) = \alpha\sqrt{\frac{m-1}{\alpha^2 + 2\alpha + m}}$$

and

$$\sigma_{\mathcal{H}}(\tilde{A}) = \alpha\sqrt{\frac{m-1}{m(\alpha^2 + 2\alpha + m)}}.$$

Proof. See Section A.2. □

For the matrix $A \in \mathbb{R}^{m\times m}$ in (4.1) with $\alpha > 0$ and $\beta = 0$, we can immediately get

$$\|A^{(1)}\|^2 = \alpha^2 + 2\alpha + m, \quad \|A^{(i)}\|^2 = m \quad (i = 2, 3, \ldots, m),$$

and

$$\|A\|_F^2 = \sum_{i=1}^{m}\|A^{(i)}\|^2 = \alpha^2 + 2\alpha + m^2.$$

Besides, from (4.3) and (4.4) we have

$$\sigma^2_{\min}(A) = \frac{1}{2}\left(\alpha^2 + 2\alpha + m^2 - \sqrt{(\alpha^2 + 2\alpha + m^2)^2 - 4(m-1)^2\alpha^2}\right)$$

and

$$\sigma^2_{\min}(\tilde{A}) = \frac{1}{2}\left(m - \sqrt{m^2 - \frac{4(m-1)^2\alpha^2}{m(\alpha^2 + 2\alpha + m)}}\right),$$

while from the definitions of the quantities γ and δ given in Theorem 3.1 (3) and (5) we get

$$\gamma = \|A\|_F^2 - \min_{1\le i\le m}\|A^{(i)}\|^2 = \alpha^2 + 2\alpha + m^2 - m$$

and

$$\delta = \max_{1\le i\le m}\|A^{(i)}\|^2 = \alpha^2 + 2\alpha + m.$$

Therefore, in accordance with the definitions of the BMRF's in (3.17) we know that

$$\begin{aligned} \mathrm{BMRF}^{\mathrm{RKU}}(\alpha) \triangleq \mathrm{BMRF}^{\mathrm{RKU}} &= 1 - \frac{\sigma^2_{\min}(\tilde{A})}{m} \\ &= 1 - \frac{1}{2}\left(1 - \sqrt{1 - \frac{4(m-1)^2\alpha^2}{m^3(\alpha^2 + 2\alpha + m)}}\right), \end{aligned} \tag{4.5}$$

$$\begin{aligned} \mathrm{BMRF}^{\mathrm{RKNU}}(\alpha) \triangleq \mathrm{BMRF}^{\mathrm{RKNU}} &= 1 - \frac{\sigma^2_{\min}(A)}{\|A\|_F^2} \\ &= 1 - \frac{1}{2}\left(1 - \sqrt{1 - \frac{4(m-1)^2\alpha^2}{(\alpha^2 + 2\alpha + m^2)^2}}\right), \end{aligned} \tag{4.6}$$

$$\mathrm{BMRF}^{\mathrm{RKR}}(\alpha) \triangleq \mathrm{BMRF}^{\mathrm{RKR}} = 1 - \frac{\sigma_{\min}^2(A)}{\gamma} = 1 - \frac{\alpha^2+2\alpha+m^2}{2(\alpha^2+2\alpha+m^2-m)}\left(1-\sqrt{1-\frac{4(m-1)^2\alpha^2}{(\alpha^2+2\alpha+m^2)^2}}\right), \tag{4.7}$$

$$\mathrm{BMRF}^{\mathrm{RKD}}(\alpha) \triangleq \mathrm{BMRF}^{\mathrm{RKD}} = 1 - \frac{\sigma_{\min}^2(\tilde{A})}{m-1} = 1 - \frac{m}{2(m-1)}\left(1-\sqrt{1-\frac{4(m-1)^2\alpha^2}{m^3(\alpha^2+2\alpha+m)}}\right), \tag{4.8}$$

$$\mathrm{BMRF}^{\mathrm{KMR}}(\alpha) \triangleq \mathrm{BMRF}^{\mathrm{KMR}} = 1 - \frac{\sigma_{\mathcal{H}}^2(A)}{\delta} = 1 - \frac{(m-1)\alpha^2}{(\alpha^2+2\alpha+m)^2},$$

$$\mathrm{BMRF}^{\mathrm{KMD}}(\alpha) \triangleq \mathrm{BMRF}^{\mathrm{KMD}} = 1 - \sigma_{\mathcal{H}}^2(\tilde{A}) = 1 - \frac{(m-1)\alpha^2}{m(\alpha^2+2\alpha+m)}.$$

The following factors play an important role in the comparisons of the BMRF's of the Kaczmarz-type methods:

$$f_1(\alpha) = \frac{m^3(\alpha^2+2\alpha+m)}{(\alpha^2+2\alpha+m^2)^2}, \quad f_2(\alpha) = \frac{\alpha^2+2\alpha+m^2}{\alpha^2+2\alpha+m^2-m}, \quad f_3(\alpha) = \frac{\alpha^2+2\alpha+m}{\alpha^2+2\alpha+m^2}.$$

In fact, by using the two positive constants

$$\alpha_1 = \sqrt{m^3-3m^2-m+1}-1 \quad \text{and} \quad \alpha_2 = \sqrt{m^3-2m^2+2m+1}-1, \tag{4.9}$$

and noticing that $\alpha_2 > \alpha_1$ for all $m \geq 1$ and $\alpha_2 > \alpha_1 > m$ for any $m \geq 5$, we can prove that these three factors have the properties as shown in the following lemma.

Lemma 4.2. *Let $\alpha > 0$ and m be a positive integer. Then the following statements hold true:*

(i) *for the function $f_1(\alpha)$, if $5 \leq m \leq \alpha \leq \alpha_1$, we have $f_1(\alpha) \geq \frac{m}{m-1}$;*
(ii) *for the function $f_2(\alpha)$, we have $1 \leq f_2(\alpha) \leq \frac{m}{m-1}$;*
(iii) *for both functions $f_1(\alpha)$ and $f_2(\alpha)$, if $m \geq 5$ and $\alpha \geq \alpha_2$, we have $f_1(\alpha)\, f_2(\alpha) \leq 1$;*
(iv) *for the function $f_3(\alpha)$, if $\alpha \leq \sqrt{m+1}-1$ and $m \geq 6$, we have $f_3(\alpha) \leq \frac{1}{\sqrt{2m}}$, and if $\alpha \geq m \geq 5$, we have $f_3(\alpha) \geq \frac{1}{\sqrt{m-1}}$.*

Proof. See Section A.3. □

Based upon Lemma 4.2, we can qualitatively compare the quantities $\mathrm{BMRF}^{\#}$, with $\#$ = RKU, RKNU, RKR, RKD, and KMR. The corresponding results are precisely stated in the following theorem.

Theorem 4.2. *Let $\alpha > 0$, and m be a positive integer satisfying $m \geq 6$. Let α_1 and α_2 be two positive constants defined as in (4.9). Then the following statements hold true:*

(1) *if $\alpha \in [m, \alpha_1]$, we have*

$$\mathrm{BMRF}^{\mathrm{RKR}} \leq \mathrm{BMRF}^{\mathrm{RKNU}} \leq \mathrm{BMRF}^{\mathrm{RKD}} \leq \mathrm{BMRF}^{\mathrm{RKU}} \leq \mathrm{BMRF}^{\mathrm{KMR}}; \tag{4.10}$$

(2) *If $\alpha \in [\alpha_2, +\infty)$, we have*

$$\mathrm{BMRF}^{\mathrm{RKD}} \leq \mathrm{BMRF}^{\mathrm{RKU}} \leq \mathrm{BMRF}^{\mathrm{RKR}} \leq \mathrm{BMRF}^{\mathrm{RKNU}} \leq \mathrm{BMRF}^{\mathrm{KMR}}; \tag{4.11}$$

(3) *if $\alpha \in (0, \sqrt{m+1}-1]$, we have*

$$\mathrm{BMRF}^{\mathrm{KMR}} \leq \min\left\{\mathrm{BMRF}^{\mathrm{RKU}}, \mathrm{BMRF}^{\mathrm{RKNU}}, \mathrm{BMRF}^{\mathrm{RKR}}, \mathrm{BMRF}^{\mathrm{RKD}}\right\}. \tag{4.12}$$

Proof. See Section A.4. □

Based on Theorem 4.2, we intuitively show in Table 6 the comparable relationships among the five Kaczmarz-type methods RKU, RKNU, RKR, RKD, and KMR, in terms of BMRF.

In this table, we use the symbol "≤" or "≥" without a superscript to show that the corresponding inequality relation is valid unconditionally, while we use the symbol "≤" with one or more superscripts to show that the corresponding inequality

Table 6
The comparable relationships in terms of BMRF as shown in Theorem 4.2.

Method	RKU	RKNU	RKR	RKD	KMR
RKU		$\le^{(2)}$	$\le^{(2)}$	$\ge$	$\le^{(1),(2)}$
RKNU	$\le^{(1)}$		$\ge$	$\le^{(1)}$	$\le^{(1),(2)}$
RKR	$\le^{(1)}$	$\le$		$\le^{(1)}$	$\le^{(1),(2)}$
RKD	$\le$	$\le^{(2)}$	$\le^{(2)}$		$\le^{(1),(2)}$
KMR	$\le^{(3)}$	$\le^{(3)}$	$\le^{(3)}$	$\le^{(3)}$	

Table 7
The intuitive skeleton for the comparable relationships in terms of BMRF as shown in Table 6.

Method	RKU	RKNU	RKR	RKD	KMR
RKU		–	–	$\ge$	–
RKNU			$\ge$	–	–
RKR				–	–
RKD					–
KMR					

relation is valid only conditionally. For example, "$\le^{(1)}$" with the superscript '(1)' means that the corresponding inequality relation holds true under the condition (1) in Theorem 4.2, and "$\le^{(1),(2)}$" with the superscripts '(1), (2)' means that the corresponding inequality relation holds true only under the conditions (1) and (2) in Theorem 4.2.

In particular, we see from Table 6 that $\mathrm{BMRF}^{\mathrm{RKU}} \ge \mathrm{BMRF}^{\mathrm{RKNU}}$ is valid under the condition (1), and $\mathrm{BMRF}^{\mathrm{RKU}} \le \mathrm{BMRF}^{\mathrm{RKNU}}$ is valid under the condition (2). Consequently, whether or not $\mathrm{BMRF}^{\mathrm{RKU}}$ and $\mathrm{BMRF}^{\mathrm{RKNU}}$ are comparable is problem-dependent.

More briefly, in Table 7 we exhibit the comparable relationships among these five Kaczmarz-type methods, which have been demonstrated in Theorem 3.2 (ii) and Theorem 4.2. This table is also a simplified and skeleton version of Table 6. In Table 7, We still use the symbol "$\ge$" to denote that the value of BMRF in that row is greater than the value of BMRF in the corresponding column, or in other words, the method in that column has a smaller BMRF than the method in the corresponding row. Besides, we also use the symbol "–" to denote that which BMRF is smaller is non-deterministic and problem-dependent.

From Table 7, we observe that the comparable relationships among BMRF's of these five Kaczmarz-type methods are problem-dependent, except for the deterministic comparative relationship between $\mathrm{BMRF}^{\mathrm{RKR}}$ and $\mathrm{BMRF}^{\mathrm{RKNU}}$, and between $\mathrm{BMRF}^{\mathrm{RKD}}$ and $\mathrm{BMRF}^{\mathrm{RKU}}$, since Theorem 3.2 (ii) guarantees that $\mathrm{BMRF}^{\mathrm{RKR}} \le \mathrm{BMRF}^{\mathrm{RKNU}}$ and $\mathrm{BMRF}^{\mathrm{RKD}} \le \mathrm{BMRF}^{\mathrm{RKU}}$.

Admittedly, Theorem 3.2 (ii)-(iii) and Theorem 4.2 only provides a qualitative comparison among the BMRF's of the six Kaczmarz-type methods RKU, RKNU, RKR, RKD, KMR, and KMD, which is not capable of showing clearly how much the BMRF of one of these methods is either smaller or larger than the BMRF of another. In Figs. 1 and 2, we further give a quantitative exhibition about the relative differences and varying tendency, in which we plot the curve of ratio versus α in Fig. 1 and the curve of BMRF versus α in Fig. 2, when $m = 30$ and $m = 50$, respectively, for the above-mentioned six Kaczmarz-type methods. Here the term "ratio" is defined as the ratio of $\mathrm{BMRF}^{\mathrm{KMD}}$ against BMRF's of the other five methods RKU, RKNU, RKR, RKD, and KMR, that is,

$$\mathrm{ratio} = \frac{\min\left\{\mathrm{BMRF}^{\mathrm{RKU}}, \mathrm{BMRF}^{\mathrm{RKNU}}, \mathrm{BMRF}^{\mathrm{RKR}}, \mathrm{BMRF}^{\mathrm{RKD}}, \mathrm{BMRF}^{\mathrm{KMR}}\right\}}{\mathrm{BMRF}^{\mathrm{KMD}}}.$$

As a matter of fact, Theorem 3.2 (iii) has shown that the MD rule is the best among the six selection rules U, NU, R, D, MR, and MD, in terms of BMRF's of their corresponding Kaczmarz-type methods.

From Fig. 1, we see that the ratio is greater than 1 for all values of α. Moreover, it is monotonically increasing with respect to the parameter α, and the increasing rate is significantly large especially when α is large enough. This provides a numerical confirmation for Statement (iii) of Theorem 3.2, that is, the MD rule is the best among the six selection rules.

From Fig. 2, we observe that the difference between $\mathrm{BMRF}^{\mathrm{RKR}}$ and $\mathrm{BMRF}^{\mathrm{RKNU}}$ is very small and almost neglectable, though $\mathrm{BMRF}^{\mathrm{RKR}} < \mathrm{BMRF}^{\mathrm{RKNU}}$ is guaranteed in general by Theorem 3.2 (ii). The same observation is applicable to $\mathrm{BMRF}^{\mathrm{RKD}}$ and $\mathrm{BMRF}^{\mathrm{RKU}}$. When $m = 30$, among these methods, in terms of BMRF we see that KMR is the best when α is small, RKR is the best when α belongs to the interval (30, 150), and RKD is the best when α is larger than 160. Moreover, the value of $\mathrm{BMRF}^{\mathrm{RKD}}$ changes slightly with respect to the parameter α, while those of both $\mathrm{BMRF}^{\mathrm{RKR}}$ and $\mathrm{BMRF}^{\mathrm{KMR}}$ change distinctly with respect to the parameter α especially when α is not significantly large. The reason for causing such a numerical phenomenon may be that the smallest nonzero singular value of the matrix $\tilde{A}$, that is, $\sigma_{\min}(\tilde{A})$, is hardly influenced by the parameter α if all of the row norms of the matrix $\tilde{A}$ is normalized to be 1.

In summary, based on the theoretical results in Theorem 3.2, Theorem 4.1, and Theorem 4.2, we can obtain the following synthetic conclusions about the comparable relationships of the Kaczmarz-type methods in terms of MSD_k and BMRF, which are intuitively expressed in Table 8.

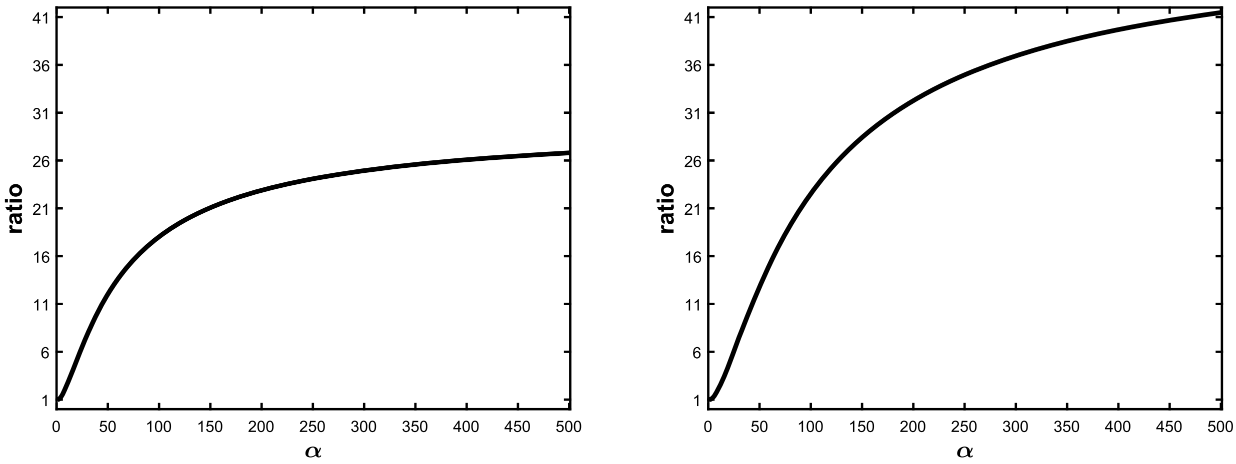


Fig. 1. Pictures of ratio versus the parameter α for $\alpha \in (0, 500]$: $m = 30$ (left) and $m = 50$ (right).

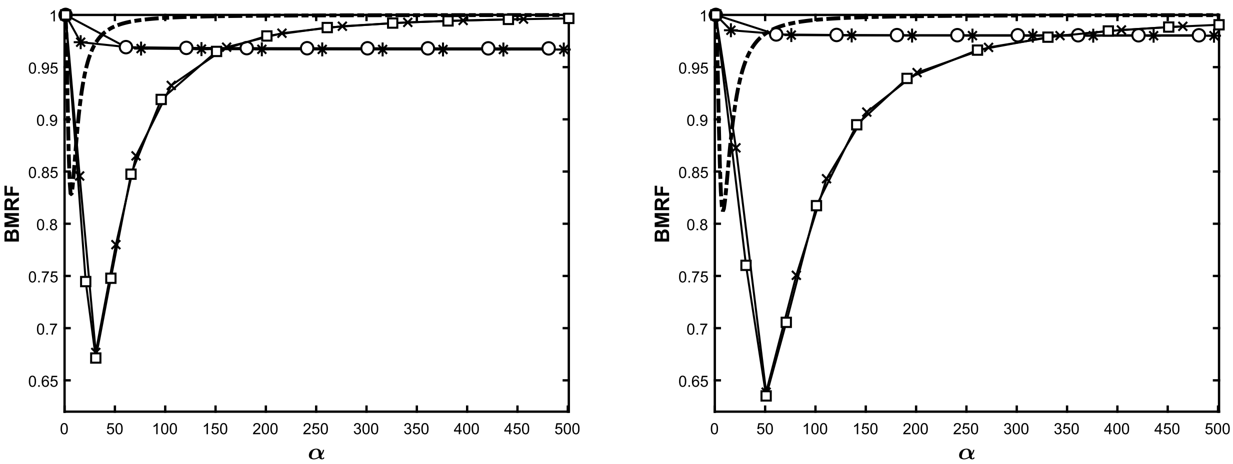


Fig. 2. Pictures of BMRF versus the parameter α for $\alpha \in (0, 500]$ with different Kaczmarz-type methods: $m = 30$ (left) and $m = 50$ (right). RKU: "$- \circ -$", RKNU: "$- \times -$", RKR: "$-\square-$", RKD: "$- * -$", and KMR: "$- \cdot -$".

Table 8
The synthetic comparable relationships for the selection rules of working rows in the Kaczmarz-type methods.

Rule	U	NU	R	D	MR	MD
U						
NU	P					
R	◯	B				
D	B	◯	P			
MR	P	◯	P	P		
MD	B	B	B	B	B	

(i) The R rule is better than the NU rule, the D rule is better than the U rule, and the MD rule is the best among all of these selection rules. This conclusion is symbolized with the letter "B" in Table 8.
(ii) The comparable relationships between the selection rules R and U, D and NU, and MR and NU have not been established yet in either theory or experiments. This conclusion is symbolized with the notation "◯" in Table 8.
(iii) The comparable relationships among the other selection rules are problem-dependent. This conclusion is symbolized with the letter "P" in Table 8.

5. Numerical experiments

In this section, we implement the six Kaczmarz-type methods RKU, RKNU, RKR, RKD, KMR, and KMD, and assess their numerical performance by solving the linear system (1.1) in the real vector and matrix spaces from a large set of problems, and we employ the number of iteration steps, briefly indicated as "IT", as the evaluation indicator, where the IT represents the median of the numbers of iteration steps with respect to 50-time repeated runs of the corresponding iteration method.

The right-hand side $b \in \mathbb{R}^m$ of the linear system (1.1) is set to be $Ax_\star$, that is, $b = Ax_\star$, where $x_\star \in \mathbb{R}^n$ is a vector generated randomly by the MATLAB function `randn`. All iteration processes are started from the initial guess $x_0 = 0$, and terminated once the current iterate, say, e.g., x_k, satisfies the condition

$$\mathrm{RSE} \triangleq \frac{\|x_k - x_*\|^2}{\|x_0 - x_*\|^2} < 10^{-5},$$

where $x_* = A^\dagger b$, with $A^\dagger$ being the Moore-Penrose generalized inverse of the matrix $A \in \mathbb{R}^{m\times n}$ computed by the MATLAB function `pinv`. Here we remark that RSE stands for the *relative solution error* at the k-th iteration step of a specified iteration method.

The numerical experiments are implemented in MATLAB (version R2017b), and executed on a personal computer with 3.60 GHz central processing unit (Intel(R) Core(TM) i7-7700), 16.0 GB of RAM, and Windows 10 operating system.

Example 5.1. Consider the linear system (1.1) with $n = m$ and its coefficient matrix $A \in \mathbb{R}^{m\times m}$ being of the form (4.1) in Section 4. We let $m = 50$.

In Fig. 3, we plot the curves of log(RSE) versus IT for the six Kaczmarz-type methods, in which the left-top picture corresponds to the case $\alpha = -8$ and $\beta = 10$, the right-top picture corresponds to the case $\alpha = 100$ and $\beta = 10$, the left-bottom picture corresponds to the case $\alpha = -8$ and $\beta = 0$, and the right-bottom picture corresponds to the case $\alpha = 100$ and $\beta = 0$. Here, the horizontal axis, i.e., IT, represents the number of iteration steps, while the vertical axis, i.e., log(RSE), represents the base-10 logarithm of RSE, with both of them being the corresponding medians with respect to 50-time repeated runs of the referred method.

If $\beta \neq 0$, then the coefficient matrix $A \in \mathbb{R}^{m\times m}$ of the linear system (1.1) is of full rank. For this case, Fig. 3 (a)-(b) show the following phenomena:

(i) When the absolute values of α and β are approximately equal, that is, $|\alpha| \approx |\beta|$, then the row norms of the coefficient matrix A are about the same, and RKU and RKNU, RKR and RKD, and KMR and KMD exhibit a similar numerical performance, respectively; See Remark 3.3.

(ii) When the absolute value of α is far away from that of β, then the row norms of the coefficient matrix A are significantly different. For this case, the RKU method performs much better than the RKNU method.

If $\beta = 0$ but $\alpha \neq 0$, then the coefficient matrix $A \in \mathbb{R}^{m\times m}$ of the linear system (1.1) is of rank 2. For this case, Fig. 3 (c)-(d) show that the RKR, RKD, KMR, and KMD methods can achieve the terminating criterion very quickly, while the other two methods, i.e., RKU and RKNU, converge very slowly, with the convergence speed of RKU being even much slower than that of RKNU.

Example 5.2. Let $m = 500$ and $n = 50$, and $A_{\mathrm{rand}} \in \mathbb{R}^{m\times n}$ be a matrix generated randomly by the MATLAB function `randn`. Then we consider the linear system (1.1) with its coefficient matrix $A \in \mathbb{R}^{m\times n}$ being reconstructed from the random matrix A_{rand} according to the following manners:

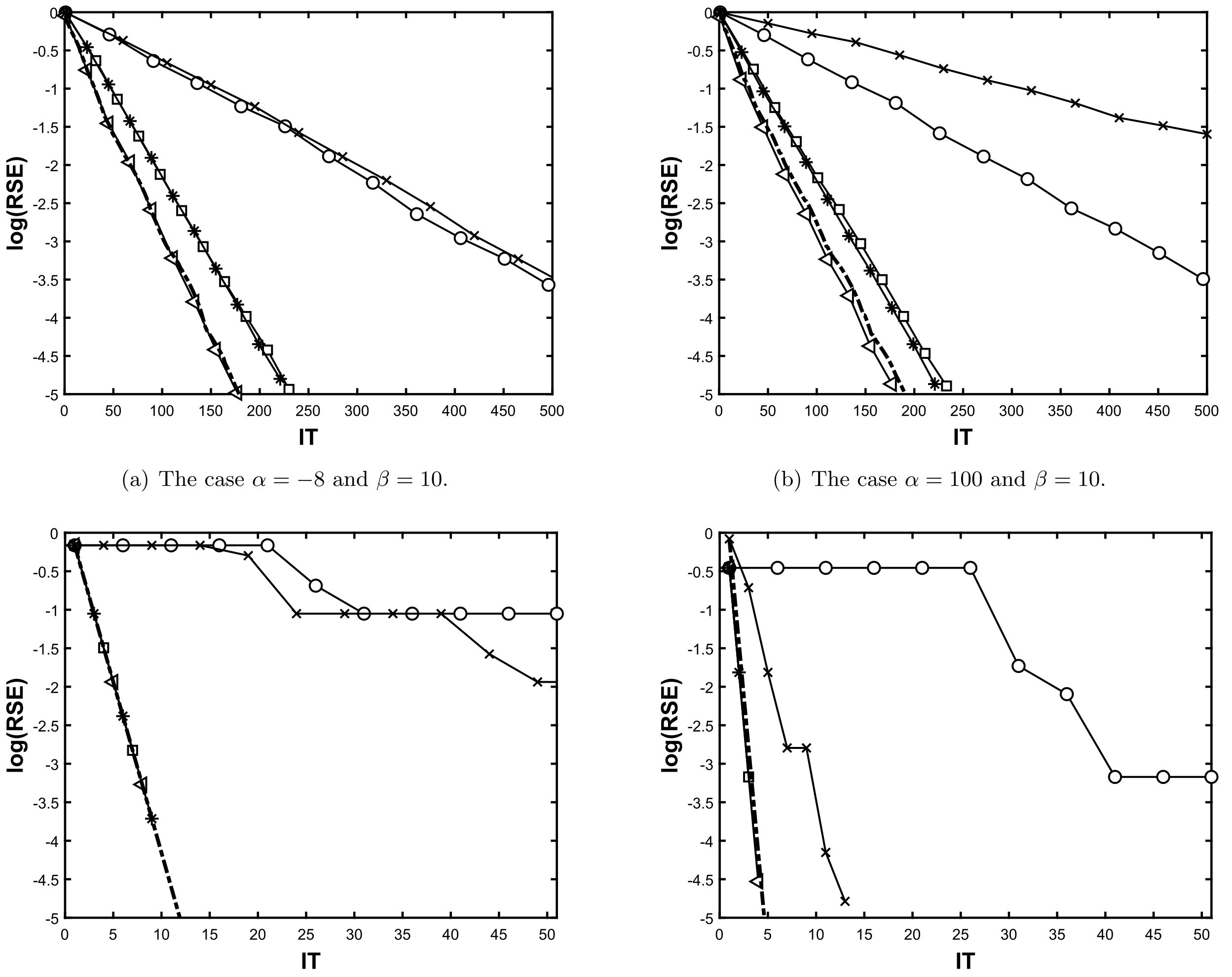
(i) The matrices A and A_{rand} have the same rows except that we choose 50 rows randomly from the matrix A_{rand} and then multiply each of these rows with the constant 10, resulting in the corresponding 50 rows of the matrix A, which are modified from the matrix A_{rand}.

(ii) Let i_* be the index of the row having the maximum row norm in the matrix A_{rand}, $i_l = \max\{i_* - 25, 1\}$, and $i_r = \min\{i_* + 25, m\}$. Then the matrices A and A_{rand} have the same rows except that for the rows from i_l to i_r of the matrix A_{rand} we multiply each of them with the constant 5, resulting in the corresponding $i_r - i_l + 1$ rows of the matrix A, which are modified from the matrix A_{rand}.

(iii) The matrices A and A_{rand} have the same rows except that for the rows from $100\ell - 99$ to $100\ell - 50$ of the matrix A_{rand} we multiply each of them with the constant $6\ell - 3$, $\ell = 1, 2, 3, 4, 5$, resulting in the corresponding 50×5 rows of the matrix A, which are modified from the matrix A_{rand}.

(iv) Let $m_o = \frac{m}{10} = 50$, and the $m = 500$ rows of the matrix A_{rand} be evenly divided into 10 parts in their natural ordering as

$$\mathrm{rowset}_i = \{i \times m_o + \ell \mid \ell = 1, 2, \ldots, m_o\}, \quad i = 0, 1, \ldots, 9.$$



(a) The case $\alpha = -8$ and $\beta = 10$.

(b) The case $\alpha = 100$ and $\beta = 10$.

(c) The case $\alpha = -8$ and $\beta = 0$.

(d) The case $\alpha = 100$ and $\beta = 0$.

Fig. 3. Pictures of log(RSE) versus IT for the Kaczmarz-type methods with respect to the linear system (1.1) in Example 5.1. RKU: "$-\circ-$", RKNU: "$-\times-$", RKR: "$-\square-$", RKD: "$-*-$", KMR: "$-\cdot-$", and KMD: "$-\triangleleft-$".

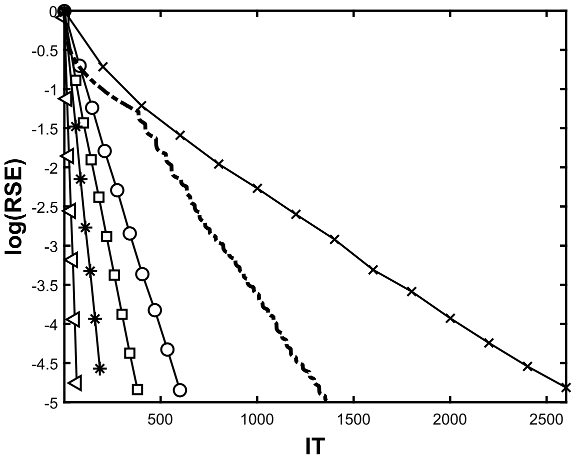
Then the matrix A is obtained by multiplying all of the rows of the matrix A_{rand} in the set rowset$_i$ by a constant randomly generated by the MATLAB function `randn`.

Note that the matrix $A \in \mathbb{R}^{m\times n}$ constructed in such manners possesses significantly different distributions of the row norms, though the matrix $A_{\text{rand}} \in \mathbb{R}^{m\times n}$ may not do.

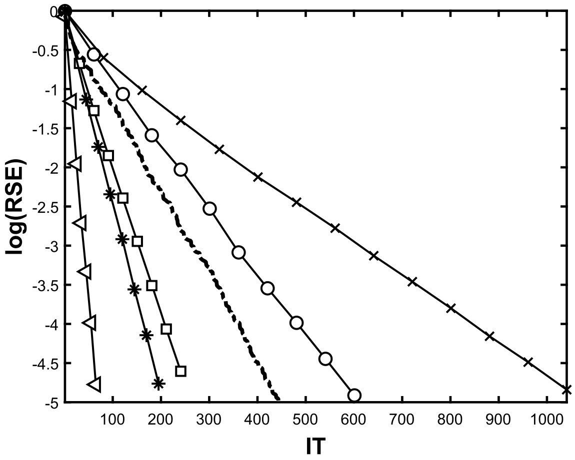
In Fig. 4, we plot the curves of log(RSE) versus IT for the six Kaczmarz-type methods with respect to the matrices described in Example 5.2, in which the left-top picture corresponds to Case (i), the right-top picture corresponds to Case (ii), the left-bottom picture corresponds to Case (iii), and the right-bottom picture corresponds to Case (iv). Here again, the horizontal axis, i.e., IT, represents the number of iteration steps, while the vertical axis, i.e., log(RSE), represents the base-10 logarithm of RSE, with both of them being the corresponding medians with respect to 50-time repeated runs of the referred method.

From Fig. 4, we see that the numerical performance of the KMR method undulates drastically, as sometimes it performs better than the RKU, RKR, and RKD methods but sometimes it performs worse than these three methods. Besides, Fig. 4 also shows that the RKU method is greatly advantageous over the RKNU method in terms of the iteration counts, when the coefficient matrix $A \in \mathbb{R}^{m\times n}$ of the linear system (1.1) has a few rows whose row norms are far greater than the others.

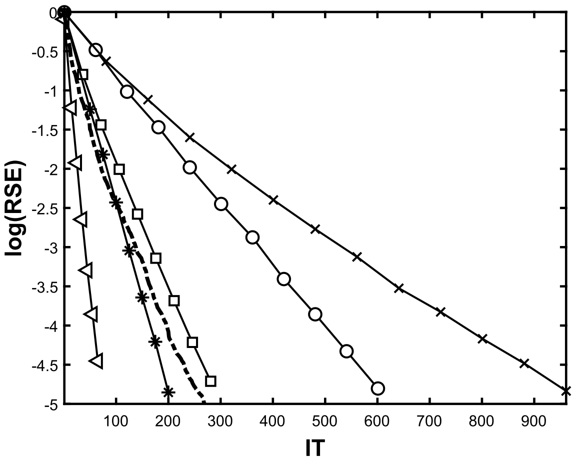
Example 5.3. Consider the linear system (1.1) with its coefficient matrix $A \in \mathbb{R}^{m\times n}$ being the following sparse matrices: model1, GL7d26, Trefethen_300, GD96_a, abtaha1, and GL7d12, which are chosen from [13].



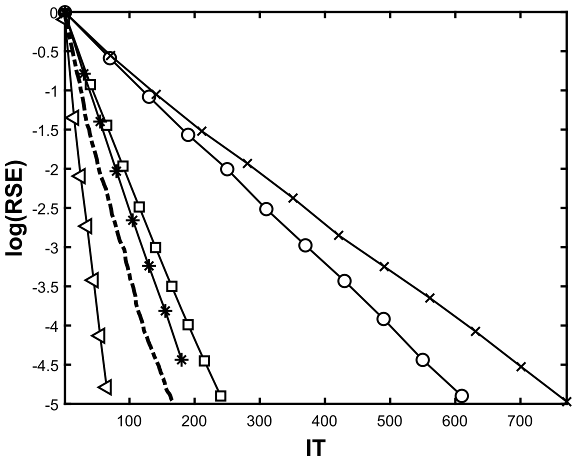
(a) The matrix A in Case (i)



(b) The matrix A in Case (ii)



(c) The matrix A in Case (iii)



(d) The matrix A in Case (iv)

Fig. 4. Pictures of log(RSE) versus IT for the Kaczmarz-type methods with respect to the linear system (1.1) in Example 5.2. RKU: "$-\circ-$", RKNU: "$-\times-$", RKR: "$-\square-$", RKD: "$-*-$", KMR: "$-\cdot-$", and KMD: "$-\triangleleft-$".

Table 9
IT of the Kaczmarz-type methods for the sparse matrices $A \in \mathbb{R}^{m\times n}$ in Example 5.3.

Name	model1	GL7d26	Trefethen_300	GD96_a	abtaha1	GL7d12
$m \times n$	362×798	305×2789	300×300	1096×1096	14596×209	8899×1019
RKU	18045	3044	14219	28842	33015	118677
RKNU	41101	454270	–	21042	56623	43726
RKR	4204	909	2229	2774	5181	5189
RKD	4287	719	918	2653	3421	5661
KMR	3893	1009	1920	2178	880	2963
KMD	3608	581	758	1946	491	1975

We remark that the matrices in Example 5.3 originate in linear programming problem, combinatorial problem, and directed multigraph problem. The matrices model1 and GL7d26 are fat (i.e., $m < n$), the matrices Trefethen_300 and GD96_a are square (i.e., $m = n$), and the matrices abtaha1 and GL7d12 are thin (i.e., $m > n$). In addition, the matrices model1, Trefethen_300, and abtaha1 are of full rank, while the other matrices GL7d26, GD96_a, and GL7d12 are of deficient rank.

In Table 9, we report the numbers of iteration steps for the six Kaczmarz-type methods with respect to these sparse matrices. In this table, the symbol "–" indicates that the corresponding iteration method can not achieve the stopping criterion RSE $< 10^{-5}$ within 2×10^6 iteration steps.

From the numerical data in Table 9, we observe that the RKNU method requires less numbers of iteration steps than the RKU method for the matrices GD96_a and GL7d12, and the RKR method requires less numbers of iteration steps than the RKD method for the matrices model1 and GL7d12. This is different from the numerical phenomena shown in Fig. 4, in which we see that the RKD method performs better than the RKR method, and the RKU method performs better than the RKNU method. Moreover, for the matrices abtaha1, Trefethen_300, and GL7d26, the KMR method shows either more or less numbers of iteration steps than the RKD and RKR methods.

By comparing with the synthetic comparable relationships for the selection rules of working rows in the Kaczmarz-type methods in Table 8, we know that the numerical results in Fig. 4 and Table 9 further confirm the relationships marked with the symbol "B", that is, the MD rule is the best among all of the six selection rules, the D rule is better than the U rule, and the R rule is better than the NU rule. In addition, the numerical results in Fig. 4 also validate the relationships marked with the symbol "P" between the MR rule and the U, R, D rules, while the numerical results in Table 9 verify the relationships "P" between the NU and U rules, as well as between the D and R rules.

Up to now, with both analytical computations and numerical experiments we have anatomized, discussed, and demonstrated the convergence properties of the Kaczmarz-type methods with the six selection rules of the working rows, which confirm the validity of the comparable relationships given in Table 8.

In order to entirely assess the numerical performance of these Kaczmarz-type methods, we provide the "*performance profile*" with respect to these methods in the following example.

Example 5.4. Consider the linear system (1.1) whose coefficient matrix $A \in \mathbb{R}^{m\times n}$ forms a set of 150 test matrices according to the following manners:

(I) The subset of 50 sparse random matrices, with the dimensions being $m = 5000$ and $n = 100, 500, 1000, 2000, 3000$, and $n = 5000$ and $m = 100, 500, 1000, 2000, 3000$, respectively, are generated randomly by using the MATLAB function `sprandn`, in which for each fixed dimension we generate 5 matrices of the density being equal to $0.005, 0.01, 0.05, 0.2, 0.5$. Here the density of a matrix is defined as

$$\text{density} = \frac{\text{the number of nonzero entries of an } m \times n \text{ matrix}}{mn}.$$

(II) The subset of 50 sparse random matrices, with certain modifications of some rows of the random matrix A_{rand} that is generated by the MATLAB function `sprandn` with the dimensions being the same as in (I) and the density being equal to 0.05. Set $\tilde{m} = \min\{\frac{m}{10}, n\}$. Then these modification strategies are described in detail as follows:

(i) Choose $\tilde{m}$ positive integers from the set $\{1, 2, \ldots, m\}$ by making use of the MATLAB function `unidrnd`, and multiply the constant 10 to those rows of the matrix A_{rand} with their row indices corresponding to the chosen $\tilde{m}$ positive integers.

(ii) Let i_* be the index of the row having the maximum row norm in the matrix A_{rand}, $i_l = \max\{i_* - \frac{\tilde{m}}{2}, 1\}$, and $i_r = \min\{i_* + \frac{\tilde{m}}{2}, m\}$. Then the matrices A and A_{rand} have the same rows except that for the rows from i_l to i_r of the matrix A_{rand} we multiply each of them with the constant 10, resulting in the corresponding $i_r - i_l + 1$ rows of the matrix A, which are modified from the matrix A_{rand}.

(iii) The matrices A and A_{rand} have the same rows except that for the rows from $(2i-1)\tilde{m}+1$ to $2i\tilde{m}$ of the matrix A_{rand} we multiply each of them first with the constant 10 and then with a random number generated by the MATLAB function `randn`, $i = 1, 2, \ldots, \frac{m}{2\tilde{m}}$, resulting in the corresponding rows of the matrix A, which are modified from the matrix A_{rand}.

(iv) The matrices A and A_{rand} have the same rows except that for the rows from $(2i-1)\tilde{m}+1$ to $2i\tilde{m}$ of the matrix A_{rand} we multiply each of them with the constant $i+1$, $i = 1, 2, \ldots, \frac{m}{2\tilde{m}}$, resulting in the corresponding rows of the matrix A, which are modified from the matrix A_{rand}.

(v) Choose a positive integer, denoted as i_o, from the set $\{1, 2, \ldots, m\}$ by making use of the MATLAB function `unidrnd`, and let $i_l = \max\{i_o - \frac{\tilde{m}}{2}, 1\}$, $i_r = \min\{i_o + \frac{\tilde{m}}{2}, m\}$. Then the matrices A and A_{rand} have the same rows except that for the rows from i_l to i_r of the matrix A_{rand} we divide each of them by the constant 10, resulting in the corresponding $i_r - i_l + 1$ rows of the matrix A, which are modified from the matrix A_{rand}.

(III) The subset of 50 sparse matrices chosen from [13], of which 48% is fat, 20% is square, and 32% is thin. For these matrices, the number of rows ranges from 29 to 21924, and the number of columns ranges from 9 to 28016; half of them are of full rank, and the others are of deficient rank; their densities are different ranging from 0.05% to 53.04%; and their row norms have different distributions.

We remark that the matrices $A \in \mathbb{R}^{m\times n}$ in Subset (II) have largely different distributions of row norms, and those in Subset (III) originate in diverse applications such as the linear programming, weighted bipartite graph, directed weighted graph, combinatorial problem, least-squares problem, and structural problem.

We plot the performance profile for these 150 test matrices in Fig. 5, which shows that the probability that the KMD method can give the best performance is nearly 0.86, while the probability that the KMR method can give the best performance is nearly 0.14. This is reasonable as the KMR method may perform much worse than the KMD method for a number of test problems, though the former shows similar performance to the latter in some cases. In addition, we can see that for the KMR method the minimum τ such that $\varrho_s(\tau) = 1$ is about 3.6, while for the RKR and RKD methods the minima are

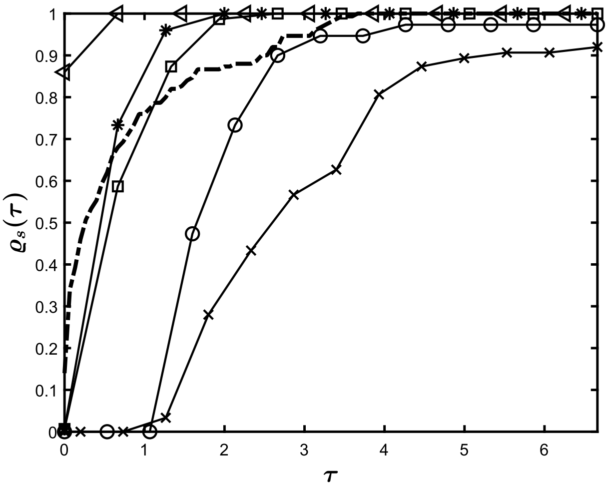


Fig. 5. Picture of $\varrho_s(\tau)$ versus $\tau \in [0, \xi]$ for the Kaczmarz-type methods with respect to the linear system (1.1) in Example 5.4. RKU: "$-\circ-$", RKNU: "$-\times-$", RKR: "$-\square-$", RKD: "$-*-$", KMR: "$-\cdot-$", and KMD: "$-\triangleleft-$".

about 2.6 and 2.0, respectively. This phenomenon demonstrates that, in general, the RKD method can solve these 150 test problems successfully and more effectively than both RKR and KMR methods. Furthermore, from Fig. 5 we also observe that the RKU and RKNU methods perform worse than the other four methods, and even they can not solve a number of the test problems within 2×10^6 iteration steps as corresponding to them it holds that $\varrho_s(\xi) < 1$; see (5.1) for the definition of ξ.

In fact, the performance profile is an intuitive and powerful tool for evaluating the computational performance of a set of numerical methods based on a set of test problems; see [1,14].

Assume that $\mathcal{S}$ is a solver set containing the six Kaczmarz-type methods, and $\mathcal{P}$ is a problem set with n_p test problems. Let $\mathcal{M}$ be a performance measure of a solver, and $\mathcal{M}_{p,s}$ denotes the measure result for the problem p using the solver s. Here we use the median number of iteration steps, i.e., IT, as the performance measure. Then we can define the best performance result for all of these six solvers on the problem p by

$$\mathcal{M}_{p,\min} = \min \left\{ \mathcal{M}_{p,s} \mid s \in \mathcal{S} \text{ and } p \text{ can be solved by } s \right\}.$$

Here, that a problem can be solved means that the corresponding IT is smaller than 2×10^6. Furthermore, we define the performance ratio as

$$\mu_{p,s} = \begin{cases} \frac{\mathcal{M}_{p,s}}{\mathcal{M}_{p,\min}}, & \text{if } p \text{ can be solved by } s, \\ \mu_\infty, & \text{if } p \text{ can not be solved by } s, \end{cases}$$

where μ_∞ is a sufficiently larger constant. Evidently, the smaller the $\mu_{p,s}$ is, the more effective the solver s is for solving the problem p.

Let

$$\xi = \max \left\{ \ln \mu_{p,s} \mid s \in \mathcal{S},\ p \in \mathcal{P} \text{ and } p \text{ can be solved by } s \right\}. \tag{5.1}$$

Then we define the *performance profile* of a solver s as

$$\varrho_s(\tau) = \frac{1}{n_p} |\Omega_s^\tau|,$$

with

$$\Omega_s^\tau = \{ p \in \mathcal{P} \mid \ln \mu_{p,s} \le \tau \} \quad \text{and} \quad \tau \in [0, \xi].$$

Here with respect to a set we use the symbol $|\cdot|$ to denote the number of entries in this set, that is, the cardinality of this set.

In fact, the performance profile $\varrho_s(\tau)$ shows the probability that problems can be solved effectively by the solver s. The solver s is considered to be more effective if $\varrho_s(\tau)$ is bigger with smaller τ. In particular, $\varrho_s(0)$ represents the probability that the solver s can solve problems with the best performance and $\varrho_s(\xi)$ represents the probability that the solver s can solve problems successfully.

6. Concluding remarks

For the six typical selection rules U, NU, R, D, MR, and MD of the working rows and their correspondingly resulted Kaczmarz-type methods RKU, RKNU, RKR, RKD, KMR, and KMD used for iteratively solving the linear system (1.1), we have reviewed the existing upper bounds for the convergence rates of the RKU, RKNU, and RKR, and then further derived more accurate upper bounds for the convergence rates of the KMR, KMD, and RKD. In terms of both mean-squared distance (MSD) and bound of mean-squared root-convergence factor (BMRF), we have given qualitative and quantitative comparisons, discussions, and analyses among the theoretical convergence property and numerical computation behavior of these Kaczmarz-type methods.

It can be concluded that KMD is the best, RKD and RKR are better than RKU and RKNU, and KMR is as good as KMD, in terms of the theoretical guarantee per iteration. However, KMR is worse than RKD and RKR, or it is even worse than RKU in some cases. In particular, if the coefficient matrix $A \in \mathbb{C}^{m\times n}$ of the linear system (1.1) has a few rows whose norms are far greater than the other rows, then RKU shows significant advantage over the RKNU.

Besides, we have found from Table 8 that for these selection rules of the working rows adopted in the Kaczmarz-type methods there are still some comparison relationships that have not been verified either in theoretical analysis or in numerical experiments, which may stem from the intrinsic gaps existing between the definitions of MSD, BMRF and the actual computational performance of the iteration methods. Therefore, it may be necessary to introduce more appropriate evaluation criterions to veritably measure and objectively assess the convergence property of these Kaczmarz-type methods and, based upon this, to further propose new working-row selection rules and row-projection iteration methods that can be employed to solve the linear system (1.1) much more effectively.

Acknowledgements

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Appendix A. Proofs of lemmas and theorems in Section 4

A.1. Proof of Theorem 4.1

Proof. By recalling that $r_0 = Ax_*$ and

$$d_0^{(i)} = \frac{|r_0^{(i)}|}{\|A^{(i)}\|}, \quad \text{for} \quad i = 1, 2, \ldots, m,$$

in accordance with (3.4) and (3.5) we straightforwardly have

$$\mathrm{MSD}_0 = \left(d_0^{(i_0)}\right)^2 = \frac{|r_0^{(i_0)}|^2}{\|A^{(i_0)}\|^2} \tag{A.1}$$

for the KMR and KMD methods, and

$$\mathrm{MSD}_0 = \sum_{i=1}^{m} p_i \cdot \frac{|r_0^{(i)}|^2}{\|A^{(i)}\|^2} = \sum_{i=1}^{m} p_i \left(d_0^{(i)}\right)^2 \tag{A.2}$$

for the RKU, RKNU, RKR, and RKD methods. In addition, we define

$$v^{(i)} = \frac{d_0^{(i)}}{\|A^{(i)}\|}, \quad \text{for} \quad i = 1, 2, \ldots, m,$$

and write

$$v = (v^{(1)}, v^{(2)}, \ldots, v^{(m)})^T \in \mathbb{R}^m.$$

From $b = A\mathbf{1}_m$, $x_0 = 0$ and (4.1), we have

$$r_0 = b - Ax_0 = b = A\mathbf{1}_m = (\alpha + m, \beta + m, \ldots, \beta + m)^T,$$

so that

$$|r_0^{(i)}|^2 = \begin{cases} (\alpha + m)^2, & \text{for}\, i = 1, \\ (\beta + m)^2, & \text{for}\, i \neq 1, \end{cases} \quad i = 1, 2, \ldots, m,$$

and

$$\|r_0\| = \sqrt{(\alpha+m)^2 + (m-1)(\beta+m)^2},$$

as well as

$$\|A^{(i)}\|^2 = \begin{cases} (\alpha+1)^2 + m - 1, & \text{for } i = 1, \\ (\beta+1)^2 + m - 1, & \text{for } i \neq 1, \end{cases} \quad i = 1, 2, \ldots, m,$$

and

$$\|A\|_F^2 = (\alpha+1)^2 + m - 1 + (m-1)\left[(\beta+1)^2 + m - 1\right].$$

In addition, by straightforward computations we can obtain

$$d_0^{(i)} = \begin{cases} \frac{|\alpha+m|}{\sqrt{(\alpha+1)^2+m-1}}, & \text{for } i = 1, \\ \frac{|\beta+m|}{\sqrt{(\beta+1)^2+m-1}}, & \text{for } i \neq 1, \end{cases} \quad i = 1, 2, \ldots, m, \tag{A.3}$$

and

$$\|d_0\|^2 = \frac{(\alpha+m)^2}{(\alpha+1)^2 + m - 1} + \frac{(m-1)(\beta+m)^2}{(\beta+1)^2 + m - 1},$$

as well as

$$v^{(i)} = \begin{cases} \frac{|\alpha+m|}{(\alpha+1)^2+m-1}, & \text{for } i = 1, \\ \frac{|\beta+m|}{(\beta+1)^2+m-1}, & \text{for } i \neq 1, \end{cases} \quad i = 1, 2, \ldots, m.$$

Moreover, for the RKU, RKNU, RKR, and RKD methods, from (A.2) and (A.3) we see that

$$\begin{aligned} \mathrm{MSD}_0 &= \sum_{i=1}^{m} p_i \left(d_0^{(i)}\right)^2 = p_1 \left(d_0^{(1)}\right)^2 + (1-p_1)\left(d_0^{(2)}\right)^2 \\ &= \frac{(\alpha+m)^2 p_1}{(\alpha+1)^2 + m - 1} + \frac{(\beta+m)^2(1-p_1)}{(\beta+1)^2 + m - 1}; \end{aligned} \tag{A.4}$$

and for the KMR method, from (A.1) and (A.3) we know that

$$\mathrm{MSD}_0 = \left(d_0^{(i_0)}\right)^2 = \begin{cases} \frac{(\alpha+m)^2}{(\alpha+1)^2+m-1}, & \text{for } i_0 = 1, \\ \frac{(\beta+m)^2}{(\beta+1)^2+m-1}, & \text{for } i_0 \neq 1, \end{cases} \quad i_0 \in \{1, 2, \ldots, m\}. \tag{A.5}$$

For notational simplicity, we introduce the one-variable real functions $\phi, \psi, \eta, \zeta : [-\infty, \infty) \longrightarrow [0, \infty)$ as

$$\phi(t) = (t+1)^2 + m - 1, \quad \psi(t) = (t+m)^2$$

and

$$\eta(t) = \frac{(t+m)^2}{(t+1)^2 + m - 1} = \frac{\psi(t)}{\phi(t)}, \quad \zeta(t) = \frac{(t+m)^2}{\left[(t+1)^2 + m - 1\right]^2} = \frac{\eta(t)}{\phi(t)}.$$

Then it holds that

$$\|A^{(1)}\|^2 = \phi(\alpha), \quad \|A^{(2)}\|^2 = \phi(\beta), \quad |r_0^{(1)}|^2 = \psi(\alpha), \quad |r_0^{(2)}|^2 = \psi(\beta)$$

and

$$(d_0^{(1)})^2 = \eta(\alpha), \quad (d_0^{(2)})^2 = \eta(\beta), \quad (v^{(1)})^2 = \zeta(\alpha), \quad (v^{(2)})^2 = \zeta(\beta).$$

Also, in accordance with the definitions of MSD_0 in (A.4), we have

$$\mathrm{MSD}_0^{\mathrm{RKU}} = \frac{\eta(\alpha)+(m-1)\,\eta(\beta)}{m}, \qquad \mathrm{MSD}_0^{\mathrm{RKNU}} = \frac{\psi(\alpha)+(m-1)\,\psi(\beta)}{\phi(\alpha)+(m-1)\,\phi(\beta)},$$

$$\mathrm{MSD}_0^{\mathrm{RKR}} = \frac{\psi(\alpha)\,\eta(\alpha)+(m-1)\,\psi(\beta)\,\eta(\beta)}{\psi(\alpha)+(m-1)\,\psi(\beta)}, \quad \mathrm{MSD}_0^{\mathrm{RKD}} = \frac{\eta(\alpha)^2+(m-1)\,\eta(\beta)^2}{\eta(\alpha)+(m-1)\,\eta(\beta)}$$

and, in light of the definitions of MSD_0 in (A.5), we have

$$\mathrm{MSD}_0^{\mathrm{KMR}} = \begin{cases} \eta(\alpha), & \text{if } \psi(\alpha) \geq \psi(\beta), \\ \eta(\beta), & \text{if } \psi(\alpha) \leq \psi(\beta). \end{cases}$$

By straightforward computations, we can further obtain the following results:

Table 10
The monotone property of the auxiliary functions.

Function \ Interval	$(-\infty, t_-)$	$(t_-, -m)$	$(-m, -1)$	$(-1, t_+)$	$(t_+, 0)$	$(0, \infty)$
$\phi(t)$	$\searrow$	$\searrow$	$\searrow$	$\nearrow$	$\nearrow$	$\nearrow$
$\psi(t)$	$\searrow$	$\searrow$	$\nearrow$	$\nearrow$	$\nearrow$	$\nearrow$
$\eta(t)$	$\searrow$	$\searrow$	$\nearrow$	$\nearrow$	$\nearrow$	$\searrow$
$\zeta(t)$	$\nearrow$	$\searrow$	$\nearrow$	$\nearrow$	$\searrow$	$\searrow$

(i) either of the inequalities $\mathrm{MSD}_0^{\mathrm{RKNU}} \geq \mathrm{MSD}_0^{\mathrm{RKU}}$ and $\mathrm{MSD}_0^{\mathrm{RKR}} \geq \mathrm{MSD}_0^{\mathrm{RKD}}$ holds true if and only if

$$(\phi(\alpha) - \phi(\beta))(\eta(\alpha) - \eta(\beta)) \geq 0;$$

(ii) either of the inequalities $\mathrm{MSD}_0^{\mathrm{KMR}} \geq \mathrm{MSD}_0^{\mathrm{RKD}}$, $\mathrm{MSD}_0^{\mathrm{KMR}} \geq \mathrm{MSD}_0^{\mathrm{RKR}}$, $\mathrm{MSD}_0^{\mathrm{KMR}} \geq \mathrm{MSD}_0^{\mathrm{RKU}}$, $\mathrm{MSD}_0^{\mathrm{KMR}} \geq \mathrm{MSD}_0^{\mathrm{RKNU}}$, and $\mathrm{MSD}_0^{\mathrm{RKR}} \geq \mathrm{MSD}_0^{\mathrm{RKU}}$ holds true if and only if

$$(\psi(\alpha) - \psi(\beta))(\eta(\alpha) - \eta(\beta)) \geq 0;$$

(iii) the inequality $\mathrm{MSD}_0^{\mathrm{RKD}} \geq \mathrm{MSD}_0^{\mathrm{RKNU}}$ holds true if and only if

$$(\eta(\alpha) - \eta(\beta))(\zeta(\alpha) - \zeta(\beta)) \geq 0.$$

From the results in (i)-(iii) above, we find that the comparable relationships among the Kaczmarz-type methods (i.e., RKU, RKNU, RKR, RKD, and KMR) in terms of MSD_0 are dependent on the monotonicity of the four functions $\phi(t)$, $\psi(t)$, $\eta(t)$, and $\zeta(t)$. The actual monotone property with respect to these auxiliary functions is listed in Table 10, in which the symbols "$\nearrow$" and "$\searrow$" represent, respectively, that the corresponding function is monotonically increasing and monotonically decreasing.

From Table 10, we observe that the functions $\eta(t)$ and $\zeta(t)$ have exactly the opposite monotonicity within the intervals $(-\infty, t_-)$ and $(t_+, 0)$, which implies that

$$(\eta(\alpha) - \eta(\beta))(\zeta(\alpha) - \zeta(\beta)) \leq 0$$

is valid for all $\alpha, \beta \in (-\infty, t_-)$ or $\alpha, \beta \in (t_+, 0)$. It then follows directly from the result in (iii) that

$$\mathrm{MSD}_0^{\mathrm{RKNU}} \geq \mathrm{MSD}_0^{\mathrm{RKD}}.$$

Similarly, since the functions $\psi(t)$ and $\eta(t)$ have exactly the same monotonicity within the interval $(-\infty, 0)$, we know that

$$\mathrm{MSD}_0^{\mathrm{KMR}} \geq \mathrm{MSD}_0^{\mathrm{RKR}}.$$

In addition, based on Theorem 3.2 we can further obtain the relationships

$$\mathrm{MSD}_0^{\mathrm{RKR}} \geq \mathrm{MSD}_0^{\mathrm{RKNU}} \quad \text{and} \quad \mathrm{MSD}_0^{\mathrm{RKD}} \geq \mathrm{MSD}_0^{\mathrm{RKU}}.$$

Therefore, we can achieve the comparable relationships

$$\mathrm{MSD}_0^{\mathrm{KMR}} \geq \mathrm{MSD}_0^{\mathrm{RKR}} \geq \mathrm{MSD}_0^{\mathrm{RKNU}} \geq \mathrm{MSD}_0^{\mathrm{RKD}} \geq \mathrm{MSD}_0^{\mathrm{RKU}}.$$

This then demonstrates the validity of Statement (1) in Theorem 4.1.

In an analogous fashion, by setting α and β in different monotone intervals as shown in Table 10, we can verify the validity of Statements (2)-(4) in Theorem 4.1. □

A.2. Proof of Lemma 4.1

Proof. We first demonstrate the validity of Statement (1).

For the vector $\mathbf{1} \triangleq \mathbf{1}_{m-1}$, from [2] we know that there exists an orthogonal matrix $Q_{m-1} \in \mathbb{R}^{(m-1)\times(m-1)}$ such that

$$Q_{m-1}^T \mathbf{1} = \sqrt{m-1}\, e_1,$$

where $e_1 \in \mathbb{R}^{m-1}$ is the column vector with its first entry being 1 and other entries being zero. Denote by

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & Q_{m-1} \end{bmatrix} \in \mathbb{R}^{m\times m}.$$

Then it follows from

$$A = \begin{bmatrix} \alpha+1 & \mathbf{1}^T \\ \mathbf{1} & \mathbf{1}\mathbf{1}^T \end{bmatrix} \in \mathbb{R}^{m\times m}$$

that

$$\begin{aligned} Q^T A Q &= \begin{bmatrix} 1 & 0 \\ 0 & Q_{m-1}^T \end{bmatrix} \begin{bmatrix} \alpha+1 & \mathbf{1}^T \\ \mathbf{1} & \mathbf{1}\mathbf{1}^T \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & Q_{m-1} \end{bmatrix} \\ &= \begin{bmatrix} \alpha+1 & \sqrt{m-1}\, e_1^T \\ \sqrt{m-1}\, e_1 & (m-1)\, e_1 e_1^T \end{bmatrix} \\ &= \begin{bmatrix} \alpha+1 & \sqrt{m-1} & 0 \\ \sqrt{m-1} & m-1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \triangleq \begin{bmatrix} \mathring{A} & 0 \\ 0 & 0 \end{bmatrix}, \end{aligned}$$

with

$$\mathring{A} = \begin{bmatrix} \alpha+1 & \sqrt{m-1} \\ \sqrt{m-1} & m-1 \end{bmatrix} \in \mathbb{R}^{2\times 2}.$$

In an analogous fashion, we can obtain

$$\begin{aligned} Q^T \tilde{A} Q &= Q^T D^{-1} A Q = (Q^T D^{-1} Q)\cdot(Q^T A Q) \\ &= D^{-1} \begin{bmatrix} \mathring{A} & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \mathring{D}^{-1}\mathring{A} & 0 \\ 0 & 0 \end{bmatrix} \\ &\triangleq \begin{bmatrix} \mathring{\tilde{A}} & 0 \\ 0 & 0 \end{bmatrix}, \end{aligned}$$

with

$$\mathring{D} = \begin{bmatrix} \sqrt{\alpha^2+2\alpha+m} & 0 \\ 0 & \sqrt{m} \end{bmatrix} \in \mathbb{R}^{2\times 2} \quad \text{and} \quad \mathring{\tilde{A}} = \begin{bmatrix} \frac{\alpha+1}{\sqrt{\alpha^2+2\alpha+m}} & \sqrt{\frac{m-1}{\alpha^2+2\alpha+m}} \\ \sqrt{\frac{m-1}{m}} & \frac{m-1}{\sqrt{m}} \end{bmatrix} \in \mathbb{R}^{2\times 2},$$

as well as

$$\begin{aligned} Q^T(\tilde{A}^T\tilde{A})Q &= (Q^T\tilde{A}Q)^T\cdot(Q^T\tilde{A}Q) = \begin{bmatrix} \mathring{\tilde{A}}^T\mathring{\tilde{A}} & 0 \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} \frac{(\alpha+1)^2}{\alpha^2+2\alpha+m} + \frac{m-1}{m} & \left(\frac{\alpha+1}{\alpha^2+2\alpha+m} + \frac{m-1}{m}\right)\sqrt{m-1} & 0 \\ \left(\frac{\alpha+1}{\alpha^2+2\alpha+m} + \frac{m-1}{m}\right)\sqrt{m-1} & \frac{m-1}{\alpha^2+2\alpha+m} + \frac{(m-1)^2}{m} & 0 \\ 0 & 0 & 0 \end{bmatrix}, \end{aligned} \tag{A.6}$$

with

$$\mathring{\tilde{A}}^T\mathring{\tilde{A}} = \begin{bmatrix} \frac{(\alpha+1)^2}{\alpha^2+2\alpha+m} + \frac{m-1}{m} & \left(\frac{\alpha+1}{\alpha^2+2\alpha+m} + \frac{m-1}{m}\right)\sqrt{m-1} \\ \left(\frac{\alpha+1}{\alpha^2+2\alpha+m} + \frac{m-1}{m}\right)\sqrt{m-1} & \frac{m-1}{\alpha^2+2\alpha+m} + \frac{(m-1)^2}{m} \end{bmatrix}. \tag{A.7}$$

By straightforward computations, we can obtain the two eigenvalues of the matrix $\mathring{A}$, which give exactly the nonzero eigenvalues of the matrix A; see (4.2). Since the matrix A is symmetric, the set of its singular values is the same as the set of its eigenvalues. As a result, we have also got the nonzero singular values of the matrix A; see again (4.2). As for the nonzero singular values of the matrix $\tilde{A}$, from the equality (A.6) we know that they are equal to the singular values of the matrix $\mathring{\tilde{A}}$, which are, in turn, equal to the square roots of the eigenvalues of the matrix $\mathring{\tilde{A}}^T\mathring{\tilde{A}}$ given in (A.7). Therefore, based on (A.7), with straightforward computations we can obtain the two nonzero singular values of the matrix $\tilde{A}$ via the two eigenvalues of the matrix $\mathring{\tilde{A}}^T\mathring{\tilde{A}}$.

The validity of Statement (2) follows directly from the facts that

$$\lambda_+(A) > \lambda_-(A) > 0, \quad \sigma_+(\tilde{A}) > \sigma_-(\tilde{A}) > 0$$

and

$$\lambda_+(\mathring{A}) > \lambda_-(\mathring{A}) > 0, \quad \sigma_+(\mathring{\tilde{A}}) > \sigma_-(\mathring{\tilde{A}}) > 0,$$

when $\alpha > 0$.

We now turn to verify Statement (3). To this end, we let

$$u = (u_1, u_2, \ldots, u_m)^T \in \mathcal{H}(A)$$

and

$$v_1 = A^{(1)}u = \alpha u_1 + \mathbf{1}^T u, \quad v_i = A^{(i)}u = \mathbf{1}^T u, \qquad i = 2, 3, \ldots, m.$$

Then in light of the definitions of the subspace $\mathcal{H}(A)$ in (3.1) and the restricted infinity norm $\sigma_{\mathcal{H}}(A)$ in (3.2), we know that there exists an index $i_0 \in \{1, 2, \ldots, m\}$ such that $v_{i_0} = 0$ and

$$u_2 = u_3 = \cdots = u_m = \frac{1}{m-1}(\mathbf{1}^T u - u_1).$$

Here we have used the fact $u \in \mathcal{R}(A^*)$.

If $\mathbf{1}^T u = 0$, then $v_i = 0$ for $i = 2, 3, \ldots, m$. It follows that

$$\begin{aligned}\sigma_{\mathcal{H}}^2(A) &= \inf_{u \in \mathcal{H}(A)} \frac{\|Au\|_\infty^2}{\|u\|^2} = \inf_{\mathbf{1}^T u = 0} \frac{\max\limits_{1 \le i \le m} |v_i|^2}{\|u\|^2} = \inf_{\mathbf{1}^T u = 0} \frac{|v_1|^2}{\|u\|^2} \\ &= \inf_{u_1 \in \mathbb{R}} \frac{\alpha^2 u_1^2}{u_1^2 + \frac{1}{m-1} u_1^2} = \frac{(m-1)\alpha^2}{m}.\end{aligned}$$

If $\mathbf{1}^T u \neq 0$, then it must hold $v_1 = 0$, i.e., $\mathbf{1}^T u = -\alpha u_1$ with $u_1 \neq 0$. It follows that

$$\begin{aligned}\sigma_{\mathcal{H}}^2(A) &= \inf_{u \in \mathcal{H}(A)} \frac{\|Au\|_\infty^2}{\|u\|^2} = \inf_{\substack{\mathbf{1}^T u \neq 0 \\ v_1 = 0}} \frac{\max\limits_{1 \le i \le m} |v_i|^2}{\|u\|^2} \\ &= \inf_{u_1 \neq 0} \frac{(\mathbf{1}^T u)^2}{u_1^2 + \frac{1}{m-1}(\mathbf{1}^T u - u_1)^2} = \frac{(m-1)\alpha^2}{\alpha^2 + 2\alpha + m}.\end{aligned}$$

As a consequence, it holds that

$$\sigma_{\mathcal{H}}^2(A) = \min\left\{\frac{(m-1)\alpha^2}{m}, \quad \frac{(m-1)\alpha^2}{\alpha^2 + 2\alpha + m}\right\} = \frac{(m-1)\alpha^2}{\alpha^2 + 2\alpha + m},$$

which readily implies that

$$\sigma_{\mathcal{H}}(A) = \alpha \sqrt{\frac{m-1}{\alpha^2 + 2\alpha + m}}.$$

If we replace v_i $(i = 1, 2, \ldots, m)$ with $\tilde{v}_i$ $(i = 1, 2, \ldots, m)$, where

$$\tilde{v}_1 = \tilde{A}^{(1)}u = \frac{1}{\sqrt{\alpha^2 + 2\alpha + m}} v_1$$

and

$$\tilde{v}_i = \tilde{A}^{(i)}u = \frac{1}{\sqrt{m}} v_i, \quad i = 2, 3, \ldots, m,$$

then $\sigma_{\mathcal{H}}(\tilde{A})$ can be derived in a similar fashion. □

A.3. Proof of Lemma 4.2

Proof. In order to derive the lower bound in (i) for the real function $f_1(\alpha)$ with respect to the variable α, we consider the convex quadratic function

$$g(t) = t^2 - m^2(m-3)t + m^3.$$

Straightforward computations show that $f_1(\alpha) \ge \frac{m}{m-1}$ if and only if $g(t) \le 0$ with $t \triangleq \alpha^2 + 2\alpha$.

In fact, for

$$t_1 = m^2 + 2m \quad \text{and} \quad t_2 \triangleq \alpha_1^2 + 2\alpha_1 = m^3 - 3m^2 - m,$$

if $m \geq 5$, then it holds that

$$\begin{aligned} t_2 - t_1 &= (m^3 - 3m^2 - m) - (m^2 + 2m) \\ &= m(m^2 - 4m - 3) \\ &= m[(m-2)^2 - 7] > 0, \end{aligned}$$

that is, $t_2 > t_1$, and

$$\begin{aligned} g(t_1) &= g(m^2 + 2m) \\ &= -m^2(m^3 - 2m^2 - 11m - 4) \\ &= -m^2[(m^2 - 11)(m-2) - 26] < 0, \\ g(t_2) &= g(m^3 - 3m^2 - m) \\ &= -m^2(m^2 - 4m - 1) \\ &= -m^2[(m-2)^2 - 5] < 0. \end{aligned}$$

Hence, we have $g(t) < 0$ when $t \in [t_1, t_2]$, or in other words, $f_1(\alpha) \geq \frac{m}{m-1}$ when $m \leq \alpha \leq \alpha_1$.

The bounds in (ii) for the function $f_2(\alpha)$ can be verified straightforwardly.

In order to verify the validity of the inequality in (iii), we denote again by $t \triangleq \alpha^2 + 2\alpha$. Then the inequality $f_1(\alpha)\, f_2(\alpha) \leq 1$ is equivalent to the one

$$(\alpha^2 + 2\alpha + m^2)(\alpha^2 + 2\alpha + m^2 - m) \geq m^3(\alpha^2 + 2\alpha + m),$$

or simply,

$$t^2 - m(m-1)^2 t - m^3 \geq 0,$$

which holds true if

$$t \geq \frac{1}{2}\left[m(m-1)^2 + m\sqrt{(m^2 - 2m + 1)^2 + 4m}\right] \triangleq \tilde{t}_0.$$

Let

$$\tilde{t}_2 = \alpha_2^2 + 2\alpha_2 = m(m^2 - 2m + 2) = \frac{1}{2}\left[m(m-1)^2 + m(m^2 - 2m + 3)\right].$$

Then, if $m \geq 2$, we have

$$\begin{aligned} \tilde{t}_0 &= \frac{1}{2}\left[m(m-1)^2 + m\sqrt{(m^2 - 2m + 3)^2 - 4(m^2 - 3m + 2)}\right] \\ &= \frac{1}{2}\left[m(m-1)^2 + m\sqrt{(m^2 - 2m + 3)^2 - 4(m-1)(m-2)}\right] \\ &\leq \frac{1}{2}\left[m(m-1)^2 + m(m^2 - 2m + 3)\right] \\ &= m(m^2 - 2m + 2) \\ &= \tilde{t}_2, \end{aligned}$$

that is, $\tilde{t}_2 \geq \tilde{t}_0$. It follows directly that $f_1(\alpha)\, f_2(\alpha) \leq 1$ when $\alpha \geq \alpha_2$. We remark that $\alpha_2 > \alpha_1 > m$ is valid for all $m \geq 5$.

We now turn to demonstrate the bounds in (iv) for the function $f_3(\alpha)$. For the upper bound, if $\alpha \leq \sqrt{m+1} - 1$ and $m \geq 6$, then we have

$$\begin{aligned} f_3(\alpha) &= \frac{\alpha^2 + 2\alpha + m}{\alpha^2 + 2\alpha + m^2} = 1 - \frac{m^2 - m}{\alpha^2 + 2\alpha + m^2} \\ &\leq 1 - \frac{m^2 - m}{m^2 + m} = \frac{2}{m+1} \\ &= \frac{2\sqrt{2}}{\sqrt{2m}\left(\sqrt{m} + \frac{1}{\sqrt{m}}\right)} \leq \frac{2\sqrt{2}}{\sqrt{2m}\left(\sqrt{6} + \frac{1}{\sqrt{6}}\right)} \\ &= \frac{4\sqrt{3}}{7\sqrt{2m}} < \frac{1}{\sqrt{2m}}. \end{aligned}$$

And for the lower bound, as when $\alpha \geq m$ we have

$$\begin{aligned} f_3(\alpha) &= \frac{\alpha^2+2\alpha+m}{\alpha^2+2\alpha+m^2} = 1 - \frac{m^2-m}{\alpha^2+2\alpha+m^2} \\ &\geq 1 - \frac{m^2-m}{2(m^2+m)} > \frac{1}{2}, \end{aligned}$$

it follows immediately that $f_3(\alpha) > \frac{1}{2} \geq \frac{1}{\sqrt{m-1}}$ when $m \geq 5$. $\square$

A.4. Proof of Theorem 4.2

Proof. From Theorem 3.2 (ii), we know that for any $\alpha > 0$ there holds

$$\mathrm{BMRF}^{\mathrm{RKD}}(\alpha) \leq \mathrm{BMRF}^{\mathrm{RKU}}(\alpha) \quad \text{and} \quad \mathrm{BMRF}^{\mathrm{RKR}}(\alpha) \leq \mathrm{BMRF}^{\mathrm{RKNU}}(\alpha).$$

These two relationships can also be verified straightforwardly from the formulas of the BMRF's in (4.5), (4.6), (4.7), and (4.8).

Denote by

$$\mu(\alpha) = \frac{4(m-1)^2\alpha^2}{(\alpha^2+2\alpha+m^2)^2} \quad \text{and} \quad \nu(\alpha) = \frac{4(m-1)^2\alpha^2}{m^3(\alpha^2+2\alpha+m)}.$$

If $\alpha \in [m, \alpha_1]$, then from Lemma 4.2 (i) we have

$$f_1(\alpha) \geq \frac{m}{m-1} > 1.$$

It follows from the definitions of $\mathrm{BMRF}^{\mathrm{RKD}}$, $\mathrm{BMRF}^{\mathrm{RKNU}}$, and $f_i(\alpha)$ $(i = 1, 2, 3)$ that

$$\begin{aligned} \mathrm{BMRF}^{\mathrm{RKD}}(\alpha) &= 1 - \frac{m}{2(m-1)} \left(1 - \sqrt{1 - \frac{4(m-1)^2\alpha^2}{m^3(\alpha^2+2\alpha+m)}} \right) \\ &= 1 - \frac{m}{2(m-1)} \left(1 - \sqrt{1 - \frac{4(m-1)^2\alpha^2}{(\alpha^2+2\alpha+m^2)^2} \cdot \frac{(\alpha^2+2\alpha+m^2)^2}{m^3(\alpha^2+2\alpha+m)}} \right) \\ &= 1 - \frac{m}{2(m-1)} \left(1 - \sqrt{1 - \mu(\alpha) \cdot \frac{1}{f_1(\alpha)}} \right) \\ &= 1 - \frac{m\,\mu(\alpha)}{2(m-1)\,f_1(\alpha)\left(1+\sqrt{1-\frac{\mu(\alpha)}{f_1(\alpha)}}\right)} \\ &\geq 1 - \frac{\mu(\alpha)}{2\left(1+\sqrt{1-\mu(\alpha)}\right)} \\ &= 1 - \frac{1}{2}\left(1 - \sqrt{1-\mu(\alpha)}\right) \\ &= 1 - \frac{1}{2} \left(1 - \sqrt{1 - \frac{4(m-1)^2\alpha^2}{(\alpha^2+2\alpha+m^2)^2}} \right) \\ &= \mathrm{BMRF}^{\mathrm{RKNU}}(\alpha). \end{aligned}$$

Consequently, we have

$$\mathrm{BMRF}^{\mathrm{RKR}}(\alpha) \leq \mathrm{BMRF}^{\mathrm{RKNU}}(\alpha) \leq \mathrm{BMRF}^{\mathrm{RKD}}(\alpha) \leq \mathrm{BMRF}^{\mathrm{RKU}}(\alpha). \tag{A.8}$$

In an analogous fashion, by making use of the inequality

$$\sqrt{1-\theta} \leq 1 - \frac{1}{2}\theta, \quad \forall \theta \in [0,\ 1], \tag{A.9}$$

we can obtain the following relationship between $\mathrm{BMRF}^{\mathrm{RKU}}$ and $\mathrm{BMRF}^{\mathrm{KMR}}$:

$$\begin{aligned}
\mathrm{BMRF}^{\mathrm{RKU}}(\alpha) &= 1 - \frac{1}{2}\left(1 - \sqrt{1 - \frac{4(m-1)^2\alpha^2}{m^3(\alpha^2+2\alpha+m)}}\right) \\
&\le 1 - \frac{(m-1)^2\alpha^2}{m^3(\alpha^2+2\alpha+m)} \\
&= 1 - \frac{(m-1)\alpha^2}{(\alpha^2+2\alpha+m)^2} \cdot \frac{(m-1)(\alpha^2+2\alpha+m)}{m^3} \\
&\le 1 - \frac{(m-1)\alpha^2}{(\alpha^2+2\alpha+m)^2} \\
&= \mathrm{BMRF}^{\mathrm{KMR}}(\alpha).
\end{aligned} \tag{A.10}$$

Here the last inequality is derived by using the estimate

$$\frac{(m-1)(\alpha^2+2\alpha+m)}{m^3} \ge \frac{(m-1)(m^2+3m)}{m^3} = \frac{m^3+m(2m-3)}{m^3} > 1,$$

which is true under the assumption $m \ge 6$. Hence, with the combination of (A.8) and (A.10) we have demonstrated the validity of the statement in (4.10).

Now we turn to demonstrate the validity of the statement in (4.11). Assume $\alpha \ge \alpha_2 > m \ge 5$. Then from Lemma 4.2 (iii)-(iv) we have

$$f_1(\alpha)\, f_2(\alpha) \le 1 \quad \text{and} \quad (m-1)\,[f_3(\alpha)]^2 \ge 1, \tag{A.11}$$

which also implies

$$f_1(\alpha) \le \frac{1}{f_2(\alpha)} \le 1. \tag{A.12}$$

By using the inequality in (A.9), we have

$$\begin{aligned}
\mathrm{BMRF}^{\mathrm{RKNU}}(\alpha) &= 1 - \frac{1}{2}\left(1 - \sqrt{1 - \frac{4(m-1)^2\alpha^2}{(\alpha^2+2\alpha+m^2)^2}}\right) \\
&\le 1 - \frac{(m-1)^2\alpha^2}{(\alpha^2+2\alpha+m^2)^2} \\
&= 1 - \frac{(m-1)\alpha^2}{(\alpha^2+2\alpha+m)^2} \cdot \frac{(m-1)(\alpha^2+2\alpha+m)^2}{(\alpha^2+2\alpha+m^2)^2} \\
&= 1 - \frac{(m-1)\alpha^2}{(\alpha^2+2\alpha+m)^2} \cdot (m-1)\,[f_3(\alpha)]^2 \\
&\le 1 - \frac{(m-1)\alpha^2}{(\alpha^2+2\alpha+m)^2} \\
&= \mathrm{BMRF}^{\mathrm{KMR}}(\alpha),
\end{aligned}$$

and applying the inequalities in (A.11) and (A.12), we have

$$\begin{aligned}
\mathrm{BMRF}^{\mathrm{RKR}}(\alpha) &= 1 - \frac{\alpha^2+2\alpha+m^2}{2(\alpha^2+2\alpha+m^2-m)}\left(1 - \sqrt{1 - \frac{4(m-1)^2\alpha^2}{(\alpha^2+2\alpha+m^2)^2}}\right) \\
&= 1 - \frac{1}{2}\left(1 - \sqrt{1 - \frac{4(m-1)^2\alpha^2}{m^3(\alpha^2+2\alpha+m)} \cdot \frac{m^3(\alpha^2+2\alpha+m)}{(\alpha^2+2\alpha+m^2)^2}}\right) f_2(\alpha) \\
&= 1 - \frac{1}{2}\left(1 - \sqrt{1 - \nu(\alpha)\, f_1(\alpha)}\right) f_2(\alpha) \\
&= 1 - \frac{\nu(\alpha)\, f_1(\alpha)\, f_2(\alpha)}{2\left(1 + \sqrt{1 - \nu(\alpha)\, f_1(\alpha)}\right)} \\
&\ge 1 - \frac{\nu(\alpha)}{2\left(1 + \sqrt{1 - \nu(\alpha)}\right)}
\end{aligned}$$

$$
\begin{aligned}
&= 1 - \frac{1}{2}\left(1 - \sqrt{1 - \nu(\alpha)}\right) \\
&= 1 - \frac{1}{2}\left(1 - \sqrt{1 - \frac{4(m-1)^2\alpha^2}{m^3(\alpha^2 + 2\alpha + m)}}\right) \\
&= \mathrm{BMRF}^{\mathrm{RKU}}(\alpha).
\end{aligned}
$$

As a result, we have shown the validity of the following relationships:

$$
\mathrm{BMRF}^{\mathrm{RKD}}(\alpha) \le \mathrm{BMRF}^{\mathrm{RKU}}(\alpha) \le \mathrm{BMRF}^{\mathrm{RKR}}(\alpha) \le \mathrm{BMRF}^{\mathrm{RKNU}}(\alpha) \le \mathrm{BMRF}^{\mathrm{KMR}}(\alpha),
$$

which is exactly the statement in (4.11).

In order to prove the statement in (4.12), we additionally assume that $\alpha \in (0, \sqrt{m+1} - 1]$ with $m \ge 6$. Then $2m\,[f_3(\alpha)]^2 \le 1$ is valid according to Lemma 4.2 (iv) again. By making use of the estimate $f_2(\alpha) \le \frac{m}{m-1}$ and the inequality

$$
\sqrt{1-\theta} \ge 1 - \theta, \quad \forall \theta \in [0,\ 1],
$$

we can obtain the relationships

$$
\begin{aligned}
\mathrm{BMRF}^{\mathrm{RKR}}(\alpha) &= 1 - \frac{\alpha^2 + 2\alpha + m^2}{2(\alpha^2 + 2\alpha + m^2 - m)}\left(1 - \sqrt{1 - \frac{4(m-1)^2\alpha^2}{(\alpha^2 + 2\alpha + m^2)^2}}\right) \\
&\ge 1 - \frac{\alpha^2 + 2\alpha + m^2}{2(\alpha^2 + 2\alpha + m^2 - m)} \cdot \frac{4(m-1)^2\alpha^2}{(\alpha^2 + 2\alpha + m^2)^2} \\
&= 1 - \frac{(m-1)\alpha^2}{(\alpha^2 + 2\alpha + m)^2} \cdot \frac{2(m-1)(\alpha^2 + 2\alpha + m)^2}{(\alpha^2 + 2\alpha + m^2)^2} \cdot \frac{\alpha^2 + 2\alpha + m^2}{\alpha^2 + 2\alpha + m^2 - m} \\
&= 1 - \frac{(m-1)\alpha^2}{(\alpha^2 + 2\alpha + m)^2} \cdot 2(m-1)\,[f_3(\alpha)]^2\, f_2(\alpha) \\
&\ge 1 - \frac{(m-1)\alpha^2}{(\alpha^2 + 2\alpha + m)^2} \cdot 2m\,[f_3(\alpha)]^2 \\
&\ge 1 - \frac{(m-1)\alpha^2}{(\alpha^2 + 2\alpha + m)^2} \\
&= \mathrm{BMRF}^{\mathrm{KMR}}(\alpha)
\end{aligned}
$$

and

$$
\begin{aligned}
\mathrm{BMRF}^{\mathrm{RKD}}(\alpha) &= 1 - \frac{m}{2(m-1)}\left(1 - \sqrt{1 - \frac{4(m-1)^2\alpha^2}{m^3(\alpha^2 + 2\alpha + m)}}\right) \\
&\ge 1 - \frac{m}{2(m-1)} \cdot \frac{4(m-1)^2\alpha^2}{m^3(\alpha^2 + 2\alpha + m)} \\
&= 1 - \frac{(m-1)\alpha^2}{(\alpha^2 + 2\alpha + m)^2} \cdot \frac{2(\alpha^2 + 2\alpha + m)}{m^2} \\
&\ge 1 - \frac{(m-1)\alpha^2}{(\alpha^2 + 2\alpha + m)^2} \\
&= \mathrm{BMRF}^{\mathrm{KMR}}(\alpha).
\end{aligned}
$$

We remark that the last inequality has been achieved by applying the inequality

$$
\frac{2(\alpha^2 + 2\alpha + m)}{m^2} \le \frac{4}{m} < 1, \qquad \forall m \ge 6.
$$

Therefore, it follows directly that the relationships

$$
\mathrm{BMRF}^{\mathrm{KMR}}(\alpha) \le \mathrm{BMRF}^{\mathrm{RKD}}(\alpha) \le \mathrm{BMRF}^{\mathrm{RKU}}(\alpha)
$$

and

$$
\mathrm{BMRF}^{\mathrm{KMR}}(\alpha) \le \mathrm{BMRF}^{\mathrm{RKR}}(\alpha) \le \mathrm{BMRF}^{\mathrm{RKNU}}(\alpha)
$$

are valid, which show that the statement in (4.12) holds true. □

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